

The Future of Contamination:

How Climate Change Will Exacerbate the Effects of Historical Contamination in the Bay Area

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Introduction

This report outlines the history and ongoing efforts of four contaminated sites in the San Francisco Bay Area: Bay View Hunters Point (BVHP), Treasure Island (TI), the Zeneca site in Richmond, and the AMCO site in West Oakland. Climate change processes such as sea level rise will worsen the current situations of these sites. This report dives into their contamination histories, the communities which will be affected, the projected climate change effects, and current cleanup efforts.

Background Information

The community project partner for this report, Greenaction For Health and Environmental Justice, is committed to helping communities fight against environmental injustice across the Western United States. Their mission statement and description is: “We are a multiracial grassroots organization founded by, led by, and working with low-income and working-class urban, rural, and indigenous communities to fight for environmental justice and build a clean, healthy and just future for all. We believe we can achieve environmental and social justice by working with frontline communities and building a strong grassroots movement to create real solutions that ensure our human right to a healthy and livable environment with justice for generations to come.”¹ Greenaction was founded in 1997 with the goal of helping environmental justice communities, giving “strong campaign, organizing, strategic, technical, media and networking support.”²

Greenaction’s work spans many different arenas of environmental justice. They excel in pressuring government agencies to give proper safety measures and relocation in cases of major contamination. Additionally, Greenaction maintains a strong community presence in the remediation process at contaminated sites. This includes—but is not limited to—insisting the “government must not use corporate contractors in cleanup who have been involved in fraud and/or inadequate cleanups and remediation of sites.”³ This philosophy extends towards the methods and process of remediation as well. Greenaction pursues complete in-place remediation when possible, and it works to ensure all necessary disposal of waste occurs at least-harm landfills.⁴

Currently, Greenaction is working to revoke the contractor Tetra Tech’s radiological licenses due to their corruption and inadequate testing at Hunters Point and Treasure Island.⁵ Other current

¹ “Our Story.” Greenaction for Health and Environmental Justice. Accessed May 14, 2021. <https://greenaction.org/our-story/>.

² Greenaction Website

³ Greenaction Website

⁴ Greenaction Website

⁵ Greenaction Website

contamination control efforts include their continuing support and petitioning to get contaminated sites placed on the CERCLA Superfund list.⁶

We are grateful and proud to have worked with Greenaction on this project. It was an incredible opportunity to contribute to a body of work that is so impactful, meaningful, and well-focused. Greenaction's goals of environmental justice and their process in pursuing those goals is admirable and produced a great learning experience for all involved.

Effects of Climate Change

The four case studies detailed throughout this report all face environmental injustices from their histories of contamination. In addition to these current issues, climate change will only exacerbate these injustices in both the near and far future. Sea level rise (SLR), a major symptom of climate change, has a grisly side effect which is only just now being realized.

Climate change will cause SLR; this fact is undisputed, and although the amount of SLR is a major source of debate and research, California now recommends all state departments "utilize a minimum of 3.5 feet of sea-level rise by 2050 for planning purposes."⁷ This dramatic SLR in the next few decades will inequitably affect the communities studied in this report as all four sites are near or on the water. This makes them quite susceptible to flooding from rising seas and King Tides.⁸ Additionally, SLR brings with it an oft-overlooked counterpart: groundwater rise.

When sea levels rise, "the surface of the shallow groundwater table will also rise."⁹ This will affect most of the Bay Area as "most development [in the Bay] has occurred on wetlands filled with sand, mud, and building rubble from 19th-century construction efforts, as well as alluvium, the material that washes down from surrounding watersheds."¹⁰ This form of development enables groundwater to easily seep up, putting the Bay Area at extreme risk for any consequences of groundwater rise—and there are extreme consequences.

Groundwater rise may increase liquefaction risks.¹¹ Liquefaction zones are areas where during the "violent shaking during an earthquake, the groundwater and fill mix together and essentially

⁶ Greenaction Website

⁷ California Coastal Commission, Consider Adopting "Making California's Coast Resilient to Sea Level Rise: Principles for Aligned State Action", 76 (Cal. Apr. 29, 2020). Accessed May 8, 2021.
https://slcprdwordpressstorage.blob.core.windows.net/wordpressdata/2020/04/04-29-2020_76.pdf.

⁸ "ART Bay Shoreline Flood Explorer," map, Adapting to Rising Tides, 2017, accessed May 13, 2021,
<https://explorer.adaptingtorisingtides.org/explorer>.

⁹ City of Alameda, California, The Response of the Shallow Groundwater Layer and Contaminants to Sea Level Rise, by C. L. May, et al. (n.p.: Silvestrum Climate Associates, 2020), 1.

¹⁰ Grace Mitchell Tada, "The Sea beneath Us," Bay Nature, March 25, 2019, [Page #], accessed May 13, 2021,
<https://baynature.org/article/the-sea-beneath-us/>.

¹¹ City of Alameda, California, The Response, 17.

turn into a liquid like quicksand that cannot support structures.”¹² This puts all structures at risk, especially those “not adequately anchored to the bedrock beneath the Bay fill.”¹³

Additionally, when groundwater rises, it will bring any buried contaminants in its path with it. By “flow[ing] through hazardous contaminants currently buried in the soil and carry[ing] them towards the surface,”¹⁴ rising groundwater will distribute pollutants into surface soil, contaminate fresh groundwater supplies, and pollute stormwater runoff flowing through neighborhoods and fields,¹⁵ possibly causing “unacceptable exposure levels to humans.”¹⁶ Groundwater rise can carry contaminants into homes via basements and crawlspaces beneath them, breach sewage lines, and even vaporize, allowing people to breathe them in.¹⁷

This form of contamination is especially worrisome because it can activate “contaminants [which are] no longer monitored because they are considered contained.”¹⁸ Groundwater rise will carry these contaminants into the unsuspecting communities who have been assured the contaminants have been safely contained.

Groundwater rise will neutralize the benefits of capping, a common contamination-remediation method used at the sites discussed in this report. Capping involves placing either a clean fill of a defined thickness or a hard surface—generally asphalt or concrete—over the contaminated soil, “prevent[ing] contact with the contaminated soil.”¹⁹ Groundwater rise will permeate these barriers, carrying the formerly contained contaminants into at-risk areas.

Climate change will exacerbate environmental injustices at the sites studied in this report. These sites all feature histories intimately intertwined with contamination, and climate change will render many past contamination-remediated methods futile. SLR will affect these communities both directly and indirectly, increasing flood risks and causing groundwater rise. Groundwater rise will bring its own host of issues, including destabilizing structures, putting them at risk of collapse during an earthquake, and activating contaminants buried beneath the surface, unearthing threats buried in the past.

¹² City of Alameda, 17.

¹³ City of Alameda, 17.

¹⁴ Legislative Analyst's Office, What Threat Does Sea-Level Rise Pose to California?, A. 2020 (Cal. Aug. 2020).

¹⁵ Legislative Analyst's Office, What Threat Does Sea-Level Rise Pose to California?, A. 2020 (Cal. Aug. 2020).

¹⁶ City of Alameda, California, The Response, i.

¹⁷ Laura Klivans, “Groundwater beneath Your Feet Is Rising with the Sea. It Could Bring Long-Buried Toxic Contamination with It,” KQED, December 15, 2020, accessed May 13, 2021, <https://www.kqed.org/science/1971582/groundwater-beneath-your-feet-is-rising-with-the-sea-it-could-bring-long-buried-toxins-with-it>.

¹⁸ Klivans, “Groundwater beneath.”

¹⁹ “Engineering Controls on Brownfields Information Guide: How They Work with Institutional Controls; the Most Common Types Used; and an Introduction to Costs,” United States Environmental Protection Agency, accessed May 13, 2021, <https://www.epa.gov/brownfields/engineering-controls-brownfields-information-guide-how-they-work-institutional-controls>.

Case Study 1: Bayview Hunters Point (BVHP)

Located in the southeastern corner of San Francisco, Bayview Hunters Point (BVHP) is situated between the Bay of San Francisco and Highway 1. It borders Cesar Chavez Street to the north and the city of Brisbane to the south.²⁰ Figure 1 shows BVHP's location in the San Francisco Bay Area (highlighted in yellow).²¹ It covers 5.21 square miles and has a total population of 37,917, giving it a population density of 7,275 people per square mile.²²

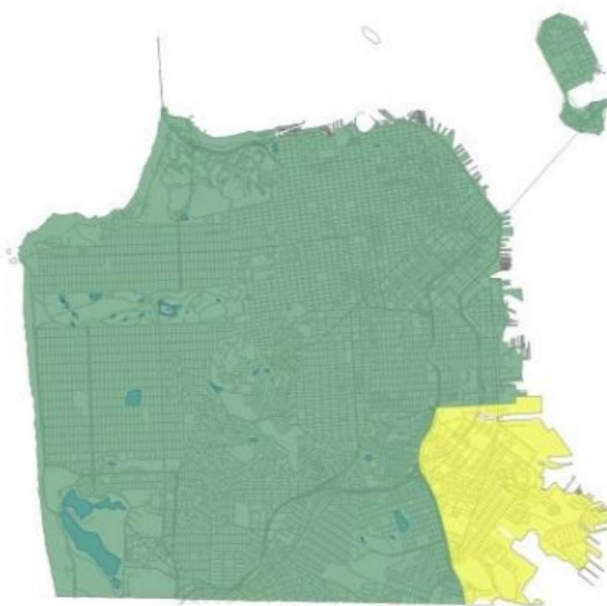


Figure 1: Location of BVHP

Community History and Overview

European settlement of BVHP began in the mid-1800s during the gold rush, and it soon gained notoriety for the plethora of butcher shops concentrated in the area. Due to its proximity to the Bay, BVHP attracted shrimp companies throughout the late 1800s and early 1900s, operations primarily led by Chinese immigrants. Hunters Point began operations as a commercial dry dock in 1869, and the United States Navy took it over in 1939. Maltese and Italian immigrants populated the area to work at the commercial dry dock, and African American laborers moved in following the transfer to the Navy.²³

²⁰ San Francisco Department of Public Health, The Bayview/Hunters Point Community Resiliency Assessment, 1.

²¹ San Francisco Department of Public Health, The Bayview/Hunters, 1.

²² U.S. Census Bureau, 2015-2019 American Community Survey 5-Year Estimates, Retrieved from <https://careshq.org/wp-content/plugins/cares-data-tools/public/views/map/tool-report.php>.

²³ "History of Bayview Hunters Point," Bayview Opera House, accessed May 7, 2021, <https://bvoh.org/culturehub/history-bvhp/>.

Due to the varied influxes of immigrants and blue-collar laborers throughout its history, BVHP hosts a highly diverse population, with 92% of the population being people of color.²⁴ Asians make up the largest percentage at 39%, followed by Black or African Americans at 25% and Hispanic or Latinos at 23%.²⁵ Figure 2 displays the demographic breakdown of BVHP as reported by the U.S. Census Bureau's 2015-2019 American Community Survey 5-Year Estimates.²⁶

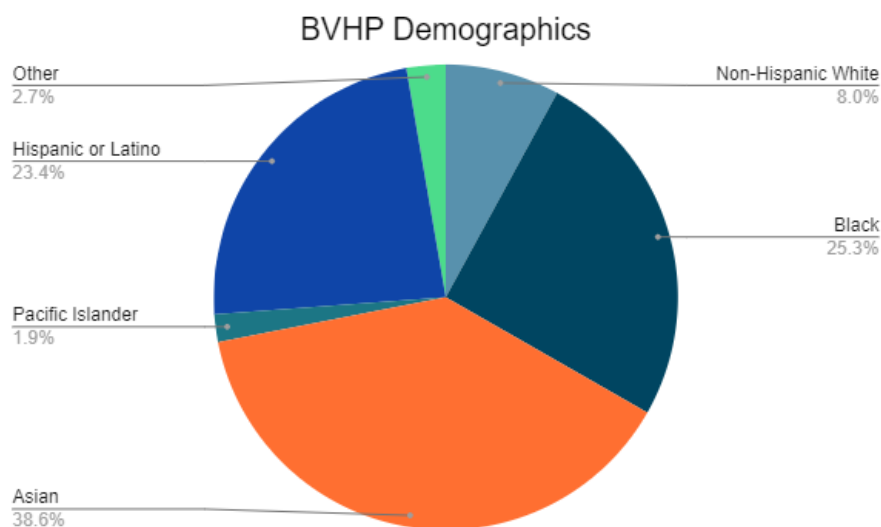


Figure 2: Population by Race/Ethnicity

BVHP has a highly vulnerable population. 35% of the population are below 200% of the Federal Poverty Level, and 51% of children below the age of 18 fall under this distinction.²⁷

Additionally, 24% of the population has no high school diploma.²⁸ These factors, combined with a milieu of others, cause most of BVHP to fall into the San Francisco Bay Conservation and Development Commission's (BCDC) top two categories for social vulnerability: High Social Vulnerability and Highest Social Vulnerability.²⁹

History of Contamination

BVHP's modern contamination issues began when the Hunters Point Shipyard (HPS) opened in 1869 as a private dry dock utilized by commercial shipping customers and the Navy. A second

²⁴ U.S. Census Bureau, American Community Survey.

²⁵ U.S. Census Bureau.

²⁶ U.S. Census Bureau.

²⁷ U.S. Census Bureau.

²⁸ U.S. Census Bureau.

²⁹ "BCDC Community Vulnerability Mapping," map, San Francisco Bay Conservation & Development Commission, <https://bcdcm.maps.arcgis.com/apps/webappviewer/index.html?id=526ca82e85eb403489de768498f605f3>.

dry dock opened in 1903, making HPS the largest drydock on the West Coast at the time.³⁰ Bethlehem Steel owned and operated HPS from 1909 until the Navy purchased it in 1939.³¹ HPS operated as a naval repair facility during World War II. In 1946,³² the Radiological Safety Section (RSS), also known as the Radiation Laboratory (RADLAB), was established at HPS “to study radiation effects, establish procedures for handling radioactive materials, and to conduct decontamination of dozens of naval vessels from nuclear tests in the Pacific.”³³ This was just the beginning of HPS’s intimate relationship with naval research on the effects of radiation, research which has left radioactive materials throughout the site.

In 1946, the U.S. Navy conducted Operation Crossroads where they detonated two atomic bombs at the Bikini Atoll in the Marshall Islands in the Pacific Ocean.³⁴ The Navy highly underestimated the severity of the radiological effects of detonating an atomic bomb underwater.³⁵ During the tests, hundreds of target ships became radiologically contaminated, and 79 of these were brought to HPS for decontamination.³⁶ The Bureau of Ships (BUSHIPS) determined decontamination materials including wet sandblast material and decontaminating solutions (such as acids) did not require special disposal.³⁷ Rather, they were simply buried at HPS or released into the harbor. In total, “a hundred or more different radionuclides...[were] brought back to or otherwise used at Hunters Point,”³⁸ contaminating the entire area.

In 1948, the RSS/RADLAB evolved into the Naval Radiological Defense Laboratory (NRDL) with the mission of engaging in “practical and applied research into the effects of radiation on living organisms and on natural and synthetic materials, in addition to continued decontamination experimentation.”³⁹ The NRDL brought copious amounts of fallout material and debris from most US nuclear tests back to HPS.⁴⁰ Additionally, they had licenses to possess and use materials such as strontium-90, yttrium-90, cobalt-60, cesium-137, plutonium-239, depleted uranium, natural thorium, and other radionuclides.⁴¹ These licenses permitted amounts of radioactive material which could contaminate trillions of tons of soil.⁴² The NRDL also conducted experiments on the effects of radiation on animals. The animal carcasses and their

³⁰ Nuclear Regulatory Commission, Historical Radiological Assessment, Volume II, Use of General Radioactive Materials, 1939-2003, Hunters Point Shipyard, Rep. (Aug. 2004), 6.2.

³¹ Nuclear Regulatory Commission, Historical Radiological Assessment, 6.2.

³² Nuclear Regulatory Commission, 6.2-4

³³ Daniel Hirsch et al., Hunters Point Naval Shipyard: The Nuclear Arms Race Comes Home, Hunters Point Publications 1 (n.p.: Committee to Bridge the Gap, 2018), 9.

³⁴ Nuclear Regulatory Commission, Historical Radiological Assessment, 6.12.

³⁵ Nuclear Regulatory Commission, 6.13.

³⁶ Hirsch et al., Hunters Point, 1-6.

³⁷ Nuclear Regulatory Commission, Historical Radiological Assessment, 6.17.

³⁸ Hirsch et al., Hunters Point, 8.

³⁹ Nuclear Regulatory Commission, Historical Radiological Assessment, 6.23.

⁴⁰ Nuclear Regulatory Commission, 6.32.

⁴¹ Hirsch et al., Hunters Point, 9.

⁴² Hirsch et al., Hunters Point, 9.

waste “were burned in an on-site incinerator,”⁴³ enabling “the potential for widespread airborne deposition of contamination.”⁴⁴ The NRDL operated at HPS until 1969.⁴⁵

The Navy decommissioned HPS in 1974⁴⁶ and leased the facilities to Triple A Machine Shop from 1976 until 1986. Triple A Machine Shop was “indicted and convicted for illegal disposal of hazardous substances,”⁴⁷ adding to HPS’s history of contamination. The Navy resumed use of HPS as a repair facility from 1986 until 1990, and in 1991, HPS was transferred to the Navy’s Base Realignment and Closure (BRAC) list before officially closing in 1994.⁴⁸

BVHP continues to experience pollution and contamination. BVHP suffers from the worst air pollution in San Francisco, with the Bay Area Air Quality Management District (BAAQMD) finding that “of 39 pollutants measured...the highest concentration—20 pollutants—was in BVHP.”⁴⁹ BVHP hosts a PG&E power plant, a sewage treatment plant which treats 80% of the solid waste generated by San Francisco, 100 Brownfield sites (a Brownfield is a property for which “the expansion, redevelopment, or reuse of which may be complicated by the presence or potential presence of a hazardous substance, pollutant, or contaminant”),⁵⁰ 124 hazardous waste handlers, and 187 leaking underground fuel tanks.⁵¹

Effects of Contamination

BVHP’s history of contamination has left copious amounts of varied contaminants in the soil. These contaminants include asbestos,⁵² which can cause asbestosis, pleural disease, lung cancer, and mesothelioma;⁵³ radioactive isotopes,⁵⁴ which can cause “bone tumors, aplastic anemia and

⁴³ Hirsch et al., Hunters Point, 15.

⁴⁴ Hirsch et al., Hunters Point, 15.

⁴⁵ Nuclear Regulatory Commission, Historical Radiological Assessment, 6.23.

⁴⁶ Department of Toxic Substances Control, “Hunters Point Naval Shipyard, Parcel A (3840001),” EnviroStor, https://www.envirostor.dtsc.ca.gov/public/profile_report?global_id=38440001.

⁴⁷ Department of Toxic Substances Control, “Hunters Point,” EnviroStor.

⁴⁸ Department of Toxic Substances Control, “Hunters Point,” EnviroStor.

⁴⁹ Aurelio Rojas, “High Rates of Disease in Bayview: Study Lends Weight to Pollution Fears,” SF Gate (San Francisco), June 9, 1997, accessed May 7, 2021, <https://www.sfgate.com/news/article/High-Rates-Of-Disease-In-Bayview-Study-lends-2837125.php>.

⁵⁰ “Overview of EPA’s Brownfields Program,” United States Environmental Protection Agency, accessed May 7, 2021, <https://www.epa.gov/brownfields/overview-epas-brownfields-program#:~:text=Definition%20of%20a%20Brownfield%20Site,substance%2C%20pollutant%2C%20or%20contaminant>.

⁵¹ Bayview Hunters Point Mothers Environmental Health & Justice Committee, Huntersview Tenants Association, and Greenaction for Health & Environmental Justice, Pollution, Health, Environmental Racism and Injustice: A Toxic Inventory of Bayview Hunters Point, San Francisco (n.p., 2004), 5.

⁵² Department of Toxic Substances Control, “Hunters Point Naval Shipyard, Parcel B & Base-Wide (38440002),” EnviroStor, accessed May 7, 2021, https://www.envirostor.dtsc.ca.gov/public/profile_report?global_id=38440002.

⁵³ U.S. Department of Health & Human Services, “Asbestos and Your Health: Health Effects of Asbestos,” Agency for Toxic Substances and Disease Registry, last modified November 3, 2016, accessed May 7, 2021, https://www.atsdr.cdc.gov/asbestos/health_effects_asbestos.html.

⁵⁴ Department of Toxic Substances Control, “Hunters Point,” EnviroStor.

myeloid metaplasia;”⁵⁵ and polychlorinated biphenyls (PCBS),⁵⁶ which can “suppress[] the immune system...reduce[] IQ and alter[] behavior...[and] increase the risk of developing cardiovascular and liver disease and diabetes.”⁵⁷ The list of contaminants and their effects is nearly boundless. In total, BVHP “contains four times as many toxins as any other city neighborhood.”⁵⁸

This milieu of toxic and harmful contaminants in the area has caused “hospitalization rates [to be] dramatically higher in BVHP in all but one category.”⁵⁹ Additionally, BVHP has “suffered a higher incidence of asthma and breast and cervical cancer than the rest of the state.”⁶⁰ Due to the environmental contamination in the area and the demographic makeup of the community, the eleven census tracts which make up BVHP are ranked between the 75th and 94th percentile for most at-risk communities by CalEnviroScreen.⁶¹ BVHP’s history of contamination puts its current residents at extreme health risks, and “on average, someone living off of Third Street in BVHP [can] expect to die 14 years sooner than someone living off of California Street in Russian Hill.”⁶²

Cleanup Efforts

In 1989, the US Environmental Protection Agency (EPA) placed HPS on the Superfund National Priorities List (NPL) and split it into seven different sites (called parcels) to facilitate cleanup efforts.⁶³ The Department of Toxic Substances Control (DTSC) was the lead agency in charge of the cleanup with the EPA providing oversight.

⁵⁵ Austin M. Brues, "Biological Hazards of Toxicity of Radioactive Isotopes," *The Journal of Clinical Investigation*, November 1, 1949, 1291, accessed May 7, 2021, <https://doi.org/10.1172/JCI102196>.

⁵⁶ Department of Toxic Substances Control, "Hunters Point," EnviroStor.

⁵⁷ David O. Carpenter, *Polychlorinated Biphenyls (PCBs): Routes of Exposure and Effects on Human Health*, research report no. 1 (Rensselaer, NY: Freund Publishing House, 2006), 1.

⁵⁸ Rojas, "High Rates."

⁵⁹ Rojas, "High Rates."

⁶⁰ Brian Howard, "Bayview Blues: San Francisco Community Activists Fight the Navy's Toxic Legacy," *Earth Action Network*, January 1, 2002, accessed May 7, 2021, <https://www.thefreelibrary.com/Bayview+blues%3A+San+Francisco+community+activists+fight+the+Navy%27s...-a082352620>.

⁶¹ "Draft CalEnviroScreen 4.0," map, California Office of Environmental Health Hazard Assessment, February 2021, accessed May 7, 2021, https://experience.arcgis.com/experience/4af93cf9888a424481d2868391af2d82/page/home/?data_id=dataSource_2-1762fb8a9ed-layer-6%3A56.

⁶² Chris Roberts, "Hunters Point Shipyard Housing: Fraud Coupled with Contamination Discoveries Fuel Doubt, Calls to Test Area," *Curbed (San Francisco)*, May 4, 2018, accessed May 7, 2021, <https://sf.curbed.com/2018/5/4/17320698/hunters-point-shipyard-safe-fraud-scandal-contamination-testing>.

⁶³ Department of Toxic Substances Control, "Hunters Point," EnviroStor.

The cleanup efforts included bioremediation, an engineered cap, soil cover, decontamination, offsite disposal, excavation, and vertical engineered barriers.⁶⁴ Nearly all these methods are susceptible to the effects of climate change and rising groundwater.

Tetra Tech Scandal

Tetra Tech was contracted to complete at least two of the parcels,⁶⁵ and they have been accused of fraud multiple times since 2011.⁶⁶ One whistleblower, Anthony Smith, claims Tetra Tech supervisors had him “replace potentially contaminated soil samples with clean soil samples...dump potentially contaminated soil into open trenches across Hunters Point...[and] sign falsified documents that were later submitted to the government.”⁶⁷ Additionally, he claims they “tampered with computer data that analyzed radiation levels.”⁶⁸ In 2018, two Tetra Tech supervisors were sentenced to federal prison for falsifying records.⁶⁹ One of them, Stephen Rolfe, stated “he was pressured by Tetra Tech’s managers to ‘get clean dirt’ because the company wasn’t interested in ‘remediating the whole goddam site.’”⁷⁰

In an internal report, Tetra Tech “acknowledged submitting false data to the Navy and EPA.”⁷¹ According to the Department of Justice, Tetra Tech submitted more than 200 false reports,⁷² and an EPA and Navy analysis indicates “at least 90% of the soil sample tests were unreliable.”⁷³ The Navy has declared Tetra Tech’s data as “not reliable”⁷⁴ and have begun “plans for retesting.”⁷⁵

⁶⁴ "Superfund Site: Hunters Point Naval Shipyard San Francisco, CA," United States Environmental Protection Agency, accessed May 7, 2021, <https://cumulis.epa.gov/supercpad/SiteProfiles/index.cfm?fuseaction=second.ous&id=0902722>.

⁶⁵ Sean Myers, "Hunters Point Shipyard Contamination, Cleanup and Development," NBC Bay Area (San Francisco), October 15, 2018, accessed May 7, 2021, <https://www.nbcbayarea.com/news/local/san-francisco-hunters-point-contamination-cleanup-d>.

⁶⁶ Chris Roberts, "Navy: Do-Over of \$250 Million Cleanup at Hunters Point Necessary," Curbed (San Francisco), January 31, 2018, accessed May 7, 2021, <https://sf.curbed.com/2018/1/31/16956458/hunters-point-toxic-cleanup-navy-responds-san-francisco>.

⁶⁷ Vicky Nguyen, Liz Wagner, and Felipe Escamilla, "Former Hunters Point Worker Claims Supervisors Ordered Him to Hide Radiation," NBC Bay Area (San Francisco), May 20, 2017, accessed May 7, 2021, <https://www.nbcbayarea.com/news/local/former-hunters-point-worker-claims-supervisors-ordered-him-to-hide-radiation/140582/>.

⁶⁸ Nguyen, Wagner, and Escamilla, "Former Hunters."

⁶⁹ Liz Wagner, "Former Tetra Tech Workers Sentenced for Falsifying Records in Hunters Point Radiation Cleanup," NBC Bay Area (San Francisco), May 3, 2018, accessed May 7, 2021, <https://www.nbcbayarea.com/news/local/former-tetra-tech-workers-sentenced-for-falsifying-records-in-hunters-point-radiation-cleanup/61577/>.

⁷⁰ Jason Fagone and Cynthia Dizikes, "US Sues Tetra Tech over Hunters Point Shipyard Work, Claiming Widespread Fraud," San Francisco Chronicle, January 15, 2019, accessed May 7, 2021, <https://www.sfchronicle.com/bayarea/article/U-S-sues-Tetra-Tech-over-Hunters-Point-shipyard-13536013.php>.

⁷¹ Greg M. Schwartz, "2020 Hindsight on Dirty Deeds Done Dirt Cheap at Hunters Point," BayView (San Francisco), October 20, 2020, accessed May 7, 2021, <https://sfbayview.com/2020/10/2020-hindsight-on-dirty-deeds-done-dirt-cheap-at-hunters-point/>.

⁷² Fagone and Dizikes, "US Sues."

⁷³ Schwartz, "2020 Hindsight."

⁷⁴ US Navy, "Former Naval Shipyard Hunters Point: Radiological Cleanup," Naval Facilities Engineering Systems Command, accessed May 7, 2021, https://www.bracpmo.navy.mil/brac_bases/california/former_shipyard_hunters_point/RadiologicalCleanup.html.

⁷⁵ US Navy, "Former Naval," Naval Facilities Engineering Systems Command.

BVHP Now

Despite the scandal of Tetra Tech's falsified data, multiple housing development projects have already been completed on the former HPS site, and many more are planned. These developments have already been the source of opposition, coverups, and lawsuits.

Two of the planned developments are the India Basin Mixed-Use Project, a combination of a six-acre parks and trails project and a development project consisting of "retail shops, apartments, and townhomes,"⁷⁶ and The Shipyard, a condominium and townhome development planned and owned by the Lennar Corporation.⁷⁷

The San Francisco Board of Supervisors approved the India Basin Mixed-Use Project in 2018 after the completion of an environmental review. However, Greenaction opposes its development due to "the Environmental Impact Report's finding that the development would result in significant, unhealthy, ongoing, unavoidable air pollution that would likely violate regulatory limits."⁷⁸

The Lennar Corporation has a history of covering up contamination. In 2018, homeowners sued Lennar and the developer Five Point Holdings for keeping "homeowners in the dark about toxic radiation in the soil."⁷⁹ Additionally, leaked emails "show a concerted effort [by Lennar Corporation and the EPA] to conceal asbestos exposures in order to avoid the shutdown of redevelopment activities."⁸⁰

BVHP has been a story of government and corporate negligence creating environmental injustice. They have put minority communities at extreme risk through contamination and development, all with a disturbing disregard for the communities' health. The communities have fought back. They have filed (and won) lawsuits.⁸¹ They have organized protests and written letters to people in power.⁸² They have formed organizations committed supporting community members.^{83,84} However, they can only do so much without official support.

⁷⁶ "India Basin Mixed-Use Project," San Francisco Planning, accessed May 7, 2021, <https://sfplanning.org/india-basin-mixed-use-project#about>.

⁷⁷ "The San Francisco Shipyard," Lennar, accessed May 7, 2021, <https://www.lennar.com/New-Homes/California/San-Francisco-Bay-Area/San-Francisco/The-San-Francisco-Shipyard>.

⁷⁸ "Bayview Hunters Point," Greenaction for Health & Environmental Justice, accessed May 7, 2021, <http://greenaction.org/bayview-hunters-point/>.

⁷⁹ "Hunters Point Shipyard Residents Reach \$6.3M Settlement in Radiation Lawsuit," CBS Bay Area (San Francisco), March 25, 2021, accessed May 7, 2021, <https://sanfrancisco.cbslocal.com/2021/03/25/hunters-point-shipyard-residents-radiation-settlement/>.

⁸⁰ Jaron Browne, "Emails Show Regulators Conspiring with Lennar to Cover up Shipyard Development Danger," Bayview (San Francisco), March 23, 2011, accessed May 7, 2021, <https://sfbayview.com/2011/03/emails-show-regulators-conspiring-with-lennar-to-cover-up-shipyard-development-danger/>.

⁸¹ "Hunters Point."

⁸² Tom Molanphy, "The Air Is Bad All over the City. It's Way Worse in Bayview Hunters Point," 48 Hills (San Francisco), September 2, 2020, accessed May 7, 2021, <https://48hills.org/2020/09/the-air-is-bad-all-over-the-city-its-way-worse-in-bayview-hunters-point/>.

⁸³ "About Us: Mission and Background," BMAGIC, accessed May 7, 2021, <http://bayviewmagic.org/about/#mission>.

⁸⁴ "What We Do," Economic Development on Third, accessed May 7, 2021, <https://www.edotbayview.org/what-we-do>.

Case Study 2: Treasure Island (TI)

Background

Treasure Island (TI) is a man-made island, located along the Bay Bridge between San Francisco and Oakland. The impetus for the creation of the island was the increasing need for an airport near San Francisco.⁸⁵ Already running out of land, they decided to fill in the shoal area near Yerba Buena Island (YBI).⁸⁶ This coincided with the additional goal of hosting a major world's fair to demonstrate San Francisco's growth as a large industrial city.⁸⁷ Immediately after the fair ended, TI began its next phase of development as a Naval Base. Until 1990, TI was held by the Navy⁸⁸.

It is crucial to understand not only the socio-cultural factors leading to the creation of TI, but also the engineering characteristics of the site. Around 29 million cubic yards of sandy fill was procured and dropped near YBI to form the base of TI.⁸⁹ This hydraulic fill process is notorious for having significant problems—"because of the slow drainage of water from the core, considerable settlement...is expected in a hydraulic-fill."⁹⁰ Additionally, "this type of construction leaves a relatively loose soil structure that is subject to liquefaction during an earthquake."⁹¹ Below the fill is a landscape varying in both sediment composition and elevation. On the north side of the island, areas overlay Young Bay Mud (YBM) deposits, while other areas sit on top of sandy sediments.⁹² Additionally, the YBM deposits sit at the deepest area of the shoal, while those closer to YBI can sit as high as 2 feet below low tide.⁹³ Finally, on top of the fill, a rock dike was built and filled in to approximately 13 feet above the low-tide level.⁹⁴ TI is known to be seismically unstable and has undergone large amounts of settlement since its initial filling.

Outside of engineering characteristics, TI is crucially important because of its widespread contamination from the Navy during its time as a training base. Above everything else, TI is notable because of the continuing, present danger that is posed to residents on the island. Radioactivity and other contaminants threaten the health and safety of these extremely vulnerable people.

⁸⁵ Stark, Kevin. "How Treasure Island Got made - and Why." <https://www.kqed.org/news/11790693/magic-city-and-the-making-of-treasure-island>.

⁸⁶ Power, Maurice, John Egan, Scott Shewbridge, John deBecker, and J. Richard Faris. "Analysis of Liquefaction-Induced Damage on Treasure Island." (Dec 19, 1997), <https://pubs.usgs.gov/pp/1551b/report.pdf>.

⁸⁷ Stark, "How Treasure Island."

⁸⁸ Stark, "How Treasure Island."

⁸⁹ USGS, "Liquefaction."

⁹⁰ Veesaert, Chris J. "Inspection of Embankment Dams." (, https://www.michigan.gov/documents/deq/wrd-dams-inspection-embankment_577395_7.pdf.

⁹¹ Veesaert, "Embankment Dams."

⁹² USGS, "Liquefaction."

⁹³ USGS, "Liquefaction."

⁹⁴ USGS, "Liquefaction."

Community Characteristics

After TI's time as a Naval Base came to a close, San Francisco saw it as an opportunity to ease housing pressures on the Bay Area. In the 1990's, TI was slated for redevelopment while some people were living in the former Naval housing units and others in newly built townhomes.⁹⁵ As part of the redevelopment process, "one-third of homes would be offered to San Francisco's homeless."⁹⁶ Contrary to popular depictions of the island as a hidden gemstone with spectacular views, many former residents remain due to the low housing costs that cannot be found anywhere else in the Bay.

Information on the TI community comes from the BCDC (Bay Conservation and Development Commission) mapping tool, based on ACS community survey data. The community as a whole is listed as having both high social vulnerability and high contamination vulnerability.⁹⁷ Compared to other areas of California, TI is in the top decile for the "Renter, No Vehicle, Single Parent, Very Low Income" categories.⁹⁸ Additionally, the sources of contamination give the census tract a CalEnviroScreen score of 89.94 (a measure of environmental vulnerabilities).⁹⁹ Although it is slightly redundant, the community also lies within the top percentile for burdens from hazardous cleanup activities.¹⁰⁰

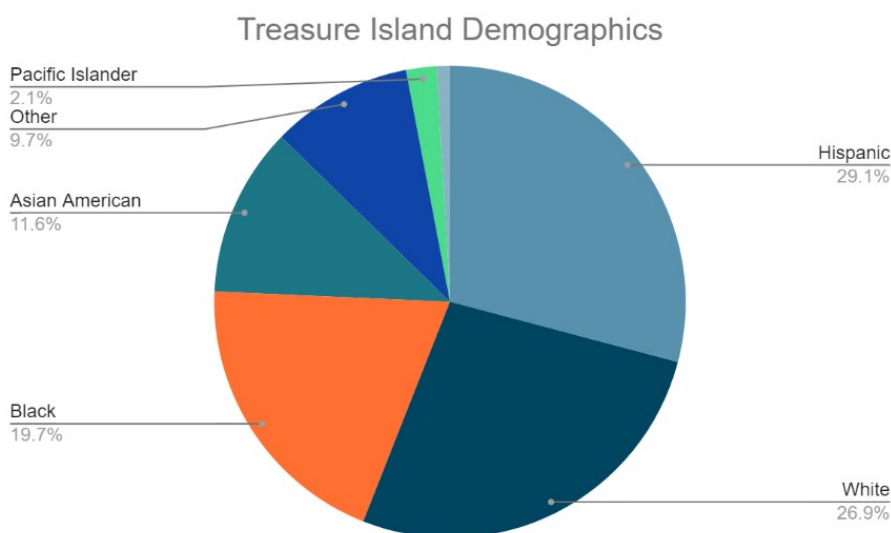


Figure 3: Treasure Island Demographics

⁹⁵ Respaut, Robin, and Reade Levinson. "Special Report: The Toxic Legacy of a CALIFORNIA Naval Base," January 31, 2019. <https://www.reuters.com/article/us-usa-military-legacy-specialreport/special-report-the-toxic-legacy-of-a-california-naval-base-idUSKCN1PP1IX>.

⁹⁶ Respaut, "Reuters."

⁹⁷ ArcGIS Web Application. Accessed May 14, 2021.

<https://bcdc.maps.arcgis.com/apps/webappviewer/index.html?id=526ca82e85eb403489de768498f605f3>.

⁹⁸ Web Application, "BCDC."

⁹⁹ Web Application, "BCDC."

¹⁰⁰ Web Application, "BCDC."

Figure 3 gives the quantitative breakdown of the community's demographics. This information comes from the ACS 2015 survey tool on the CARES website. The community is 73% people of color, with 26%, 23%, and 10% being Black, Hispanic, or Asian respectively.¹⁰¹

What makes TI unique is the confluence of factors affecting the residents. Not only are they forced onto the island by housing rates outside of their control, but they are also under constant threat from the radiation and contamination on the island. Yet, despite their danger, they face gentrification and removal from major development corporations.¹⁰² Finally, regardless of the outcome, their relationship to rising sea levels and groundwater marks them as a model for the dangers coming to the entire Bay Area in the coming years.

Seismic Hazards

As mentioned in the introduction to TI, the island is seismically questionable. The best source for studying the response of TI in a major event comes from studies conducted after the 1989 Loma Prieta earthquake. Several items are of note from those studies. First, compared with the rest of the Bay Area, the peak acceleration and shaking were initially amplified on TI due to the soft, uncompacted sand fill.¹⁰³ Observations later revealed after the initial shaking, motion died off as large parts of the island liquefied.¹⁰⁴ Additionally, many small cracks formed in the ground surface, and much of the island experienced differential settlement and lateral spreading. The areas which were most damaged were those close to the edges of the island—as would be expected from the fill process.¹⁰⁵ On the north side of the island, settlement numbers ranged from around four inches to over eight inches of settlement.¹⁰⁶

The seismic hazards on TI are numerous, and their concentration on the Northern Side of the island are further cause for concern. Settlement, liquefaction, and cracking can all exacerbate the processes that threaten greater release of contaminants.

¹⁰¹ U.S. Census Bureau (2019). 2015-2019 American Community Survey 5-Year Estimates. Retrieved from <https://careshq.org/wp-content/plugins/cares-data-tools/public/views/map/tool-report.php>.

¹⁰² Meronek, Toshio. "SF Gentrification Pushes Lower-Income Residents into Radioactive Areas." Al Jazeera America. Accessed May 14, 2021. <http://america.aljazeera.com/articles/2015/6/4/sf-gentrification-pushes-lower-income-residents-into-radioactive-areas.html>.

¹⁰³ USGS, "Liquefaction."

¹⁰⁴ USGS, "Liquefaction."

¹⁰⁵ USGS, "Liquefaction."

¹⁰⁶ USGS, "Liquefaction."

Contamination

Contamination on Treasure Island is the result of its use as part of radiological training for the US Navy.¹⁰⁷ This entire process was mirrored at the sister site at BVHP. The training schools on TI were used for decontamination training on two ships named *Independence* and *Pandemonium*.¹⁰⁸ One of those ships, the *Independence*, was shipped directly from the Bikini Atoll, fresh from its near destruction in an atomic ordinance test.¹⁰⁹ Sailors on TI used the ships as well as classrooms across the base to train on radiological cleanup procedures.¹¹⁰ While the goal may have been to teach about decontamination methods, accidents were plentiful, and the reports do not indicate safe handling.¹¹¹ Additionally, the “Treasure Island atomic warfare schools were repeatedly chastised as a result of those inspections.”¹¹² During both the initial phase of contamination as well as early cleanup efforts, the north end of the island was used as a Solid Waste Disposal Area (SWDA), with no indication as to the radiological or chemical composition of the deposited waste.¹¹³

Across the US and around the world, military bases like TI present some of the greatest contamination hazards during and after their use. Nearly two-thirds of Superfund sites are connected with military use and activity.¹¹⁴ For years, the Navy and other branches widely used industrial solvents, cleaners, and other harmful chemicals with little to no care for their toxic potentials.¹¹⁵ On TI, the sources of contamination are varied and widespread. Across the entire island there are thirty-three Comprehensive Environmental Response Compensation and Liability Act (CERCLA) sites and nine additional petroleum contaminated sites on location.¹¹⁶ These sites contain a litany of different contaminants, including radionuclides, PCBs, dioxins, TCE, lead and other heavy metals, and many others.¹¹⁷ Serious focus should be placed on what is listed as Site 12, the site closest to the area where residents live.

Within Site 12, there are many sources of contamination, but particularly the SWDAs reflect the long history of unacceptable behaviors on TI. These disposal areas contain large amounts of

¹⁰⁷ Smith, Matt, and Katherine Mieszkowski. “Treasure Island Cleanup Exposes Navy’s Mishandling of Its Nuclear Past.” Bulletin of the Atomic Scientists, June 28, 2018. <https://thebulletin.org/2014/02/treasure-island-cleanup-exposes-navys-mishandling-of-its-nuclear-past/>.

¹⁰⁸ Smith, “Cleanup Exposes.”

¹⁰⁹ Smith, “Cleanup Exposes.”

¹¹⁰ Smith, “Cleanup Exposes.”

¹¹¹ Smith, “Cleanup Exposes.”

¹¹² Smith, “Cleanup Exposes.”

¹¹³ Smith, “Cleanup Exposes.”

¹¹⁴ Hamilton, John. Contamination at U.S. Military Bases: Profiles and Responses, https://law.stanford.edu/wp-content/uploads/2017/01/hamilton_0.pdf.

¹¹⁵ Hamilton, Contamination.

¹¹⁶ “Environmental Cleanup Program Former Naval Station Treasure Island Fact Sheet.” Cal EnviroStor, November 2011. https://www.envirostor.dtsc.ca.gov/public/community_involvement/3756025016/Final%20General%20Fact%20Sheet%20No.6_11.16.11.pdf.

¹¹⁷ Fact Sheet 6, US Navy.

radionuclides with no records to give any clue as to the extent of the contamination.¹¹⁸ Excavations have revealed “Low-Level Radioactive Objects (LLROs) containing radium-226...and associated localized radium-226 contaminated soil”.¹¹⁹ Current work on the SWDAs is focused on excavation and removal, but the radionuclides are found in many more areas. Previous Greenaction project groups have mapped out the truly staggering number of LLROs found across the north side of the island.¹²⁰ In total, 1280 different objects were found up to that point, and new objects and soil clusters are being found each time excavations and surveys are undertaken.¹²¹

Community Effects

Most importantly in the case of Site 12 and its related contamination are the effects seen by the residents living on top of the waste disposal areas and next to current cleanup efforts. Interviews with young people who had experience living on the island in 2015 gives a picture of the hazards and information given to residents throughout the process.¹²² They cite the common complaint that residents were informed to not dig or plant gardens in the soil around their homes due to contamination risks.¹²³ This would seem to contrast with the consistent statements from the Navy that the area is safe for habitation. In their report, The Center for Investigative Reporting interviewed multiple residents, including a woman named Vicki Jones.¹²⁴ Jones stated, “My baby’s got a fungus in her head that the doctor can’t explain. She breaks out in bumps, but you guys don’t have answers”.¹²⁵ A Reuter’s interview with resident Kathryn Towne depicts her children finding small “treasures” in the soil near the home.¹²⁶ Later, her daughters “[suffered] rashes, asthma, thyroid issues...ovarian cysts.”¹²⁷

These experiences are representative of the possible dangers and long-term effects of living on TI for the residents. Throughout TI’s history, numerous people have suffered from likely related illnesses, from soldiers working under unsafe conditions decontaminating ships to current residents unsure if their medical woes are related to the nebulous cleanup process happening feet from their homes.¹²⁸

¹¹⁸ Work Plan, Envirostor.

¹¹⁹ Work Plan, Envirostor.

¹²⁰ Brown, Shannon, Tristan Wells, Astrian Horsburgh, and Aliya Ehde. “Treasure Island Contamination.” Google My Maps. Google. Accessed May 14, 2021. <https://www.google.com/maps/d/u/0/viewer?mid=1yS0wR9o8ZjKCrK4-SVSIrdUQ65bYrUNL&ll=37.82685797786414%2C122.37317095563799&z=16>.

¹²¹ Brown, “TI Contamination.”

¹²² Meagley, Demond, and Nanette Thompson. “The Hazards of Growing Up on Treasure Island.” KQED, July 8, 2015. <https://www.kqed.org/news/10593090/the-hazards-of-growing-up-on-treasure-island>.

¹²³ Meagley, “Youth Radio.”

¹²⁴ Smith, “Cleanup Exposes.”

¹²⁵ Smith, “Cleanup Exposes.”

¹²⁶ Respaut, “Reuters.”

¹²⁷ Respaut, “Reuters.”

¹²⁸ Respaut, “Reuters.”

Sea Level Rise

Sea Level Rise (SLR) presents a major threat to TI, the residents there, and the Bay Area as a whole. As mentioned earlier in the report, there are major dangers when contaminants are mobilized by rising sea levels and groundwater.

In particular, TI faces a second, additional danger. While lots of land in the Bay Area faces subsidence, because of TI's engineering characteristics, it has an unusually high rate of settlement.¹²⁹ This has been estimated to be around 4 millimeters a year on the North side of the island. In addition, although current development plans may seem to prepare for some SLR scenarios, they are definitely deficient, as they use only a sixteen-inch SLR projection.¹³⁰ This is less than half of the suggested 3.5-foot SLR suggested by the state of California.¹³¹ Regardless of the proposal, this low number calls into question the applicability and long-term safety of these additional geotechnical measures. Finally, the already shallow groundwater table at TI would suffer even more as the levels rise and turn towards mobilizing the radionuclides and other contaminants present on TI.

Synthesis/Suggestions

Across the board, different stories are told about Treasure Island. Some reference the location and the promise of the “jewel” in the middle of the San Francisco Bay. Development plans tout affordable housing and a futuristic recreation area for the San Francisco community. In other arenas, the history of TI is centered and explored. Callbacks to the nuclear age, WWII, and rapid expansion of the city sit as the backdrop for historical preservation on TI. Regardless, there is a deeper story at play. The residents of TI are some of the most vulnerable members of our community. Not only have they suffered and struggled to find affordable housing, they have also been forced into the dangerous and contaminated homes on TI, being given few alternatives. They are told that they should not dig or plant a garden. They are told to keep their children from playing outside. Despite these calls, they are told that it is perfectly safe, even though cover-up scandals and new radioactive objects consistently reveal themselves every year.

Above all else, this is a community in crisis. Cleanup efforts should continue outside of the control and purview of the Navy, but the residents should have their relocation paid for and provided for. It is unconscionable to leave them stranded under threat of contamination. Cleanup efforts must be thorough and varied to deal with the massive range of contamination on the island.

¹²⁹ Shirzaei, Manoochehr and Roland Bürgmann. "Global Climate Change and Local Land Subsidence Exacerbate Inundation Risk to the San Francisco Bay Area." *Science Advances* 4, no. 3 (Mar, 2018): eaap9234, <https://www.ncbi.nlm.nih.gov/pubmed/29536042>.

¹³⁰ “Geotechnical, Sea Level Rise, and Shoreline Improvements.” SF Treasure Island, n.d. https://sftreasureisland.org/sites/default/files/Final_Ch6.pdf.

¹³¹ California Coastal Commission, “Consider Adopting.”

Case Study 3: Zeneca Site in Richmond

Background

The 86-acre site is located on marsh wetlands along the San Francisco Bay shoreline in the city of Richmond, California. The address of the site is 1415 South 47th Street, Richmond, CA 94804. It is located south of Interstate 580 and east of the UC Berkeley Richmond Field Station. Historically, the site was divided into three main areas: the manufacturing plant area, the Western Research Center, and the unimproved open space area. From the late 1800s to the late 1990s, Stauffer Chemical Co., and later Zeneca Inc., manufactured industrial and agricultural chemicals including sulfuric acid and pesticides.¹³² In 1960, ponds were dug at the site to capture wastewater from industrial activities and untreated stormwater runoff. In 2002, Cherokee Simeon Ventures (CSV), a real estate company, bought the site from Zeneca Inc. to develop the property. In 2004, CSV signed deed restrictions prohibiting the construction of hospitals, homes, schools, and day care facilities on the contaminated site. The deed allowed development only for industrial or commercial use, with a distant possibility of above-ground residential units with the overseeing agency's approval (the San Francisco Regional Water Quality Board at the time).¹³³

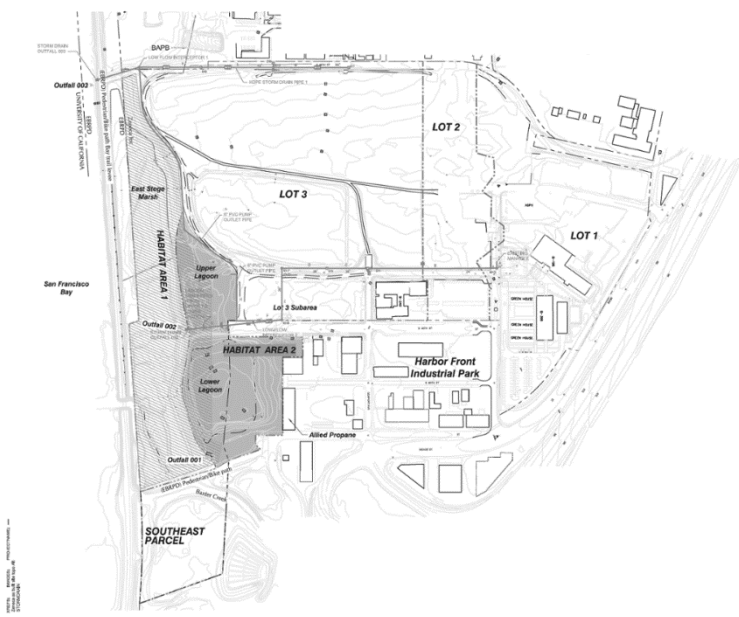


Figure 4: The site map of the Zeneca site

The site is now divided into six areas: three lots, two “habitat enhancement” areas, and the Southeast Parcel. Lot 1 which was developed into commercial space and is now known as the

¹³² State of California, “EnviroStor,” EnviroStor (Department of Toxic Substances Control), accessed May 12, 2021, https://www.envirostor.dtsc.ca.gov/public/profile_report?global_id=07280002.

¹³³ Julia Vassey, “Developing Richmond’s Zeneca Site Means Confronting a Legacy of Contamination,” Richmond Confidential, December 12, 2018, <https://richmondconfidential.org/2016/12/13/developing-richmonds-zeneca-site-means-confronting-a-legacy-of-contamination/>.

Campus Bay Business Park. Lots 2 and 3 are currently undeveloped and were zoned for light and heavy industry uses. The City of Richmond General Plan 2030 rezoned the upland portions of Lots 1, 2 and 3 to business/light industrial. Other areas were designated as open space.¹³⁴

Community Overview

The area specified for the community is the census tract where the Zeneca site is located in. The census tract has a population of about 5,931 people.¹³⁵ The population is diverse, with 27% White, 24% Black, 23% Hispanic/Latino, 18% Asian, and 5% other or multiple races. In total, the community is 70% people of color.¹³⁶



Figure 5: Census Tract 601338000

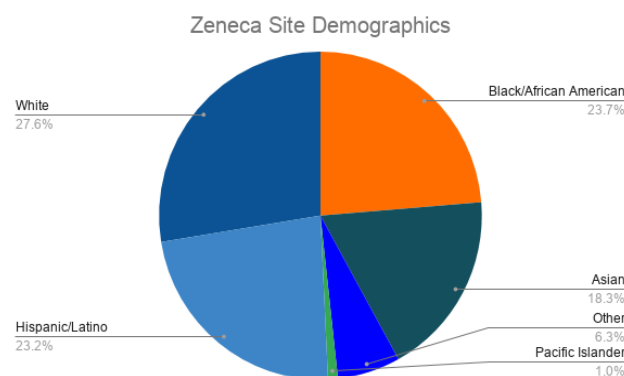


Figure 6: Zeneca Site Demographics

In terms of vulnerability, CalEnviroScreen scores the area in the 85th percentile—with 100% being the highest score.¹³⁷ The CalEnviroScreen score includes different indicators, including pollution such as waste, health indicators including asthma, and social categories such as education and poverty. The higher percentile environmental indicators in the community include diesel particulate matter, traffic, cleanup sites, groundwater threats, hazardous waste, and impaired water. These higher percentile indicators align with Interstate 580 highway and the Zeneca site being located in the community. In addition, in the social categories, 53% of the population have incomes less than 2x the poverty level, 42% of the population over 25 years old have less than a high school education, and 44% of people over the age of 16 and are eligible to work are unemployed.¹³⁸

¹³⁴ State of California, “EnviroStor.”

¹³⁵ U.S. Census Bureau, 2015-2019 American Community Survey 5-Year Estimates.

¹³⁶ U.S. Census Bureau.

¹³⁷ “CalEnviroScreen 3.0 Results,” ArcGIS Web Application (OEHHA), accessed May 12, 2021, <https://oehha.maps.arcgis.com/apps/webappviewer/index.html?id=4560cfbce7c745c299b2d0cbb07044f5>.

¹³⁸ U.S. Census Bureau.

Contamination

The past uses that caused the Zeneca site's contamination include hazardous waste storage, such as tanks and containers, illegal dumping, the manufacturing of chemicals and pesticides, weapon research, industrial treatment facility, and wastewater ponds.¹³⁹

The potential contaminants of concern at the site including metals: arsenic, beryllium and compounds, cadmium and compounds, copper and compounds, lead, mercury and compounds, nickel, selenium, and zinc, organochlorine pesticides: DDD, DDE, and DDT, polychlorinated biphenyls (PCBs), polynuclear aromatic hydrocarbons, sulfuric acid, volatile organics: benzene, tetrachlorethylene (PCE), toluene, trichloroethylene (TCE), 1,2-dichloroethylene, and vinyl chloride, carbon disulfide, EPTC, pebulate.¹⁴⁰

The potential media affected on the site includes other groundwater (uses other than drinking water), sediments, soil, and soil vapor.¹⁴¹

Although most, if not all of the contaminants have adverse health effects, the contaminants of primary concern include metals and volatile organic compounds (VOCs) which can cause breast and lung cancers, lymphoma and leukemia, and bone marrow failure.¹⁴²

Cleanup Efforts

Cleanup efforts for the Zeneca site started in 1999 after Zeneca closed the site in 1997. Zeneca was responsible for outlining and paying for the cleanup, with the San Francisco Regional Water Quality Board (RWQCB) overseeing them. This first cleanup effort extended until 2002 and included demolishing site structures and digging and treating the toxic soil. The soil was mixed with limestone to neutralize and then it was capped with a thin concrete cap, which was supposed to be a temporary measure.¹⁴³ Some of the contaminated was excavated and moved to a landfill, while some of it was treated onsite. During this time, the RWQCB rescinded Zeneca's National Pollution Discharge Elimination System permit, which let the company dump a specific amount of wastewater into the San Francisco Bay.¹⁴⁴

In 2005, the California Department of Toxic Substances Control (DTSC) took control over the cleanup effort. In 2008, a Removal Action Workplan (RAW) was approved for the Lot 1 1 PCB/VOC area. The soil in this area was found to consist of elevated levels of PCBs and VOCs

¹³⁹ State of California, "EnviroStor."

¹⁴⁰ State of California, "EnviroStor."

¹⁴¹ State of California, "EnviroStor."

¹⁴² Julia Vassey, "Developing Richmond's Zeneca Site Means Confronting a Legacy of Contamination."

¹⁴³ Janis Hashe, "The Fight to Clean Up Another Richmond Brownfield: East Bay Express: Oakland, Berkeley & Alameda," East Bay Express | Oakland, Berkeley & Alameda, July 25, 2018, <https://eastbayexpress.com/the-fight-to-clean-up-another-richmond-brownfield-2-1/>.

¹⁴⁴ Vassey, Julia. "Developing Richmond's Zeneca Site Means Confronting a Legacy of Contamination."

during the remedial investigation and was excavated.¹⁴⁵ In 2009, DTSC accused Zeneca of violating the state's Hazardous Waste Control Act for improperly treating and disposing hazardous waste, and they were fined \$500,000 as part of a settlement agreement.¹⁴⁶ Also during this year, a Time Critical Removal Action was approved to remove stockpiled soil on Lot 2 which contained high levels of pesticides and PCBs. Pilot studies for in-situ groundwater treatment of VOCs and metals, and soil vapor extraction were also conducted as part of the site investigation.¹⁴⁷

In December 2016, the Richmond City Council adopted the Richmond Bay Specific Plan (RBSP), which supports residential development along Richmond's South Shoreline, including the Zeneca site, despite the deed restrictions. Because of this, a more extensive cleanup plan is needed to clear the area of any remaining toxic materials.¹⁴⁸ However, an attorney with Edgcomb Law Group, representing Zeneca, wrote a letter to DTSC denoting that "there is no basis for the agency to require the company to remediate the entire site to accommodate a ground-floor residential use" and that it has "reservations about future homes" on the site, among other statements.¹⁴⁹ In 2017, DTSC approved a Feasibility Study/Remedial Action Plan (FS/RAP) for Habitat Area 2, which included sediment and soil excavation, groundwater monitoring and habitat restoration. In 2018, the Richmond City Council endorsed a cleanup of the site to the highest residential standard. However, in November 2019, after the developers, Shopoff Realty Investments and Hilco Redevelopment Partners, promised millions of dollars to local groups, a narrow majority of the council retracted that recommendation.¹⁵⁰ Instead, they voted to allow over 98% of the toxic materials to be left underneath the proposed high-density housing.¹⁵¹ The FS/RAP of Lot 1, Lot 2 and the uplands portion of Lot 3 were approved in 2019.¹⁵²

Discussion/Next Steps

In November 2020, Richmond voters elected a new council majority that supports a full cleanup of the site, yet that did not stop the lame duck narrow majority from voting to leave the toxic material in place and building the 4,000 housing units on top of it.¹⁵³ The city council took advantage of the COVID-19 pandemic restrictions and ignored the residents, public health and environmental experts who testified against the proposal. Due to the toxic material that will still

¹⁴⁵ State of California, "EnviroStor."

¹⁴⁶ Julia Vassey, "Developing Richmond's Zeneca Site Means Confronting a Legacy of Contamination,"

¹⁴⁷ State of California, "EnviroStor."

¹⁴⁸ Julia Vassey, "Developing Richmond's Zeneca Site Means Confronting a Legacy of Contamination,"

¹⁴⁹ Riley, "Engineering and Social Justice."

¹⁵⁰ Julia Vassey, "Developing Richmond's Zeneca Site Means Confronting a Legacy of Contamination."

¹⁵¹ Janis Hashe, "The Fight to Clean Up Another Richmond Brownfield: East Bay Express: Oakland, Berkeley & Alameda,"

¹⁵² State of California, "EnviroStor."

¹⁵³ "Lawsuit Filed to Stop Richmond Housing Development on Toxic Waste Site Owned by AstraZeneca," Citizens for East Shore Parks, January 20, 2021, <https://eastshorepark.org/lawsuit-filed-to-stop-richmond-housing-development-on-toxic-waste-site-owned-by-astrazeneca/>.

be in place, no onsite pre-schools, K-12 schools, or health or senior care facilities will be allowed on the site.¹⁵⁴

In December 2020, a coalition of Bay Area environmental and community groups, including Greenaction for Health & Environmental Justice, Sunflower Alliance, Richmond Shoreline Alliance, SPRAWLDEF, and Citizens for East Shore Parks filed a lawsuit against the City of Richmond, stating the approval of the project violates the California Environmental Quality Act (CEQA) by not considering current science on sea-level rise and more stringent health standards.¹⁵⁵

One of the reasons why some council members, including Mayor Tom Butt, voted in favor of the Zeneca project is due to much needed housing in the area, as Richmond's population is projected to increase 1.2% in the next 15 years and the demand for housing in the Bay Area is consistently rising.¹⁵⁶ Several questions come to mind in terms of this issue- yes, the Bay Area needs housing, but at what cost? The long-term health and environmental effects of the toxins would hurt the future residents and create more issues and further inequities.

¹⁵⁴ "Lawsuit Filed to Stop Richmond Housing Development on Toxic Waste Site Owned by AstraZeneca."

¹⁵⁵ "Lawsuit Filed to Stop Richmond Housing Development on Toxic Waste Site Owned by AstraZeneca."

¹⁵⁶ Egelko, Bob. "Richmond Sued over Plan to Develop Site Laden with Pollutants." San Francisco Chronicle, San Francisco Chronicle, 7 Jan. 2021, www.sfchronicle.com/bayarea/article/Richmond-sued-over-plan-to-develop-site-laden-15848909.php.

Case Study 4: AMCO Superfund Site, West Oakland

Occupied by AMCO Chemical Corp. until 1989, the AMCO Superfund Site is in the western extremum of West Oakland. The 0.9-acre site sits on the intersection of 3rd Street and Center Street, with the surrounding area a mix of residential and industrial use (Figure 8). Despite ongoing remediation efforts over the past thirty years, the site continues to release toxic waste into the environment. Residents are exposed to unsafe levels of multitude of dangerous chemicals with harmful health impacts, particularly respiratory issues.

Community

“West Oakland is a shore-front community and home to approximately 26,000 Bay Area residents from diverse economic, social, and racial backgrounds. Freeways encircle, and busy arterials and boulevards crisscross, West Oakland. Adjacent to the community is the Port of Oakland, a large and complex maritime port, with related rail yards and rail lines. West Oakland has long been a neighborhood with strong African American roots, with the community unfortunately having been shaped by redlining practices and disinvestment. Within the past decade, West Oakland has seen increased development promising to bring economic growth and jobs to the City of Oakland and the community. However, many long-time residents are concerned that new growth will increase rents and displace existing residents who can no longer afford to live in the community.”

- West Oakland Environmental Indicators Project (WOEIP)¹⁵⁷

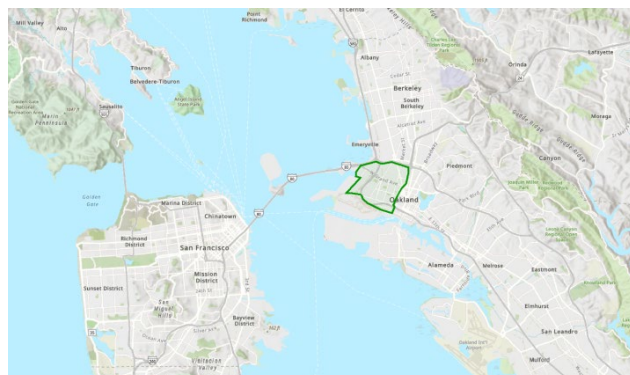


Figure 7: West Oakland, bordered in green, relative to the Bay Area



Figure 8: The AMCO Superfund Site (red) is directly adjacent to residential homes (yellow), a public park (blue) with a playground, and a homeless encampment under the 880 overpass

¹⁵⁷ West Oakland Environmental Indicators Project (WOEIP) and Bay Area Air Quality Management District, “Owning Our Air - Volume 2,” 1–1.

There is likely no group better positioned to introduce the community impacted by the AMCO site than the West Oakland Environmental Indicators Project. WOEIP is a resident-led organization started by West Oaklanders who saw the need to fight for environmental justice in their community. In partnership with the Bay Area Air Quality Management District, WOEIP documented transportation-related air pollution in West Oakland in a 2019 report, *Owning Our Air* (see Figure 9 below).¹⁵⁸

In his “Brief History of West Oakland,” archeologist and cultural historian Robert Douglass describes how “West Oakland’s fortunes had been tied to transportation from the beginning.”¹⁵⁹ From the early years of settler occupation, San Francisco used the wharf at Oakland Point (now the Port of Oakland) as its primary gateway into the interior United States. The Central Pacific Railroad completed the transcontinental railroad in 1869 and placed its terminus in Oakland. Railways forming the south and southwest borders of West Oakland attracted industrial and military operations that, along with the railways and shipyards themselves, have been polluting the air, soil, and water ever since.

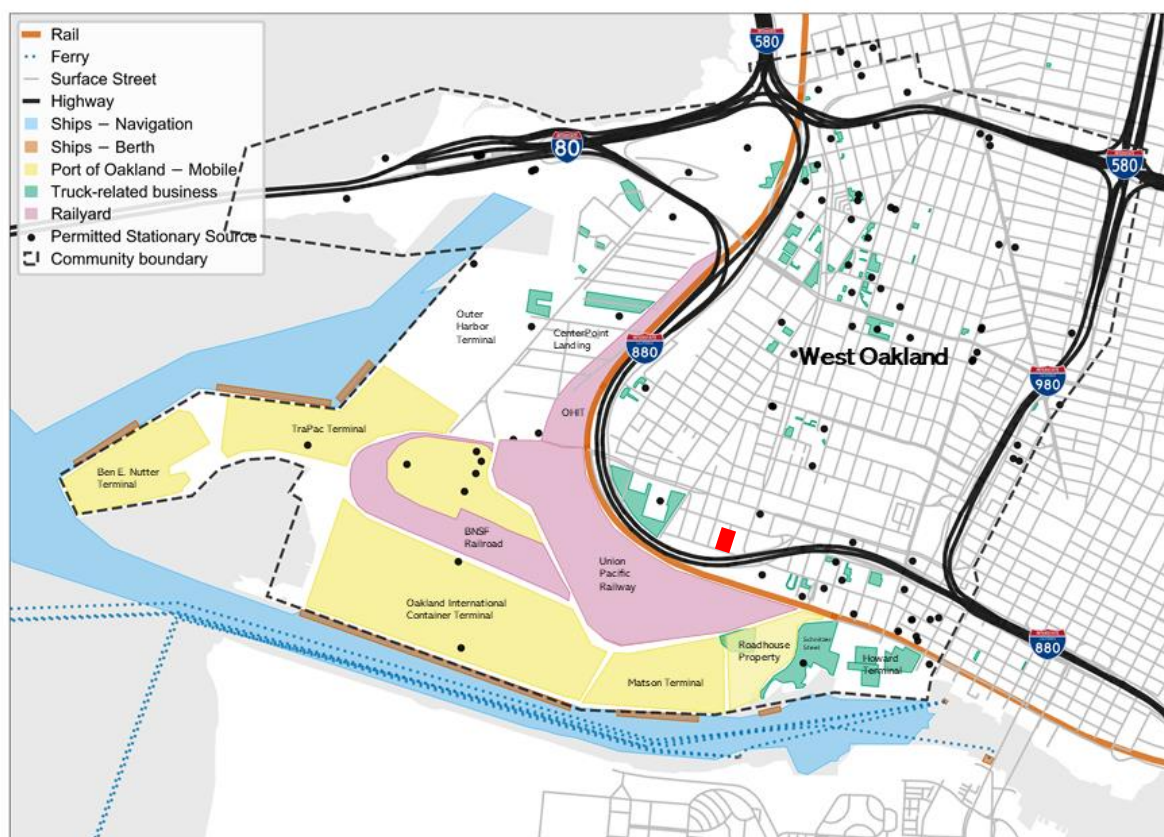


Figure 9: Transportation-related air pollution in West Oakland. The red rectangle indicates the location of the AMCO Superfund Site

¹⁵⁸ West Oakland Environmental Indicators Project (WOEIP) and Bay Area Air Quality Management District, “Owning Our Air - Volume 2.”

¹⁵⁹ Douglass, “Chapter 2: A Brief History of West Oakland.”

Oakland-based investigative journalist Alexis Madrigal compiling historical government reports, communications, and maps to expose how city planners intentionally co-located pollution sources and Black residents. The city zoned West Oakland “for heavy industry, despite its large residential populations,” but “then barred businesses (and apartment buildings) in residential districts near Lake Merritt in 1918.”¹⁶⁰ Figure 10 displays the zoning of West Oakland.¹⁶¹ City planners made this separation literally concrete with the construction of the Cypress Freeway. Completed in 1957 and destroyed in the 1989 Loma Prieta Earthquake, the Cypress Freeway cut diagonally through the heart of West Oakland, where Mandela Parkway now runs. The highway created “an infrastructural division between the mixed-race (western) and white (eastern) parts of the neighborhood.”¹⁶² In a 1937 report, Oakland’s city planning engineer, I.S. Shattuck, explicitly explains the highway’s purpose:

“The housing advantage of this diagonal highway route is that it would segregate that portion of the district under discussion where non-white families are largely concentrated from the portion where they are relatively few in number...A plan of housing could take advantage of this physical division by providing a unit or units for mixed race occupancy west of the [highway], and units for all-white occupancy east of the highway.”

– I.S. Shattuck, Engineer, Oakland City Planning Commission (1936)¹⁶³

Starting with and sustained by racist city planning and government actions, the most severe transportation-related air pollution and contamination from industrial waste exists in the intended “mixed race” western portion of West Oakland, including that emanating from the AMCO Superfund Site.

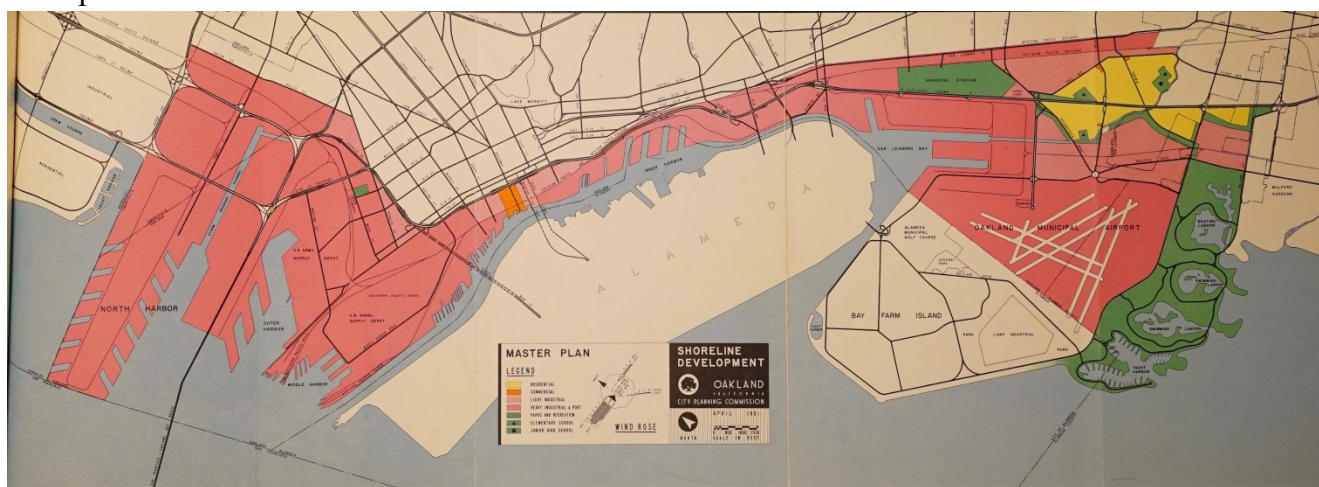


Figure 10: Zoning map showing West Oakland zoned as industrial (pink)

¹⁶⁰ Madrigal, “Oakland - How the Bay Was Built.”

¹⁶¹ Madrigal.

¹⁶² Madrigal.

¹⁶³ As cited by Madrigal in “Oakland – How the Bay Was Built.”

Demographics

The demographic (poverty and race) data presented in this section describe the seven census tracts closest to the AMCO Superfund Site (see Figure 11). Data was collected from the American Community Survey, 2015-2019 five-year estimate.¹⁶⁴

Race

Most of the roughly 10,000 people living in the census tracts closest to the AMCO site are people of color, with the largest fraction (45%) being Black residents.

Poverty

According to census data for the census tracts close to the AMCO site (bordered in red, see maps above), approximately 55% of people are living in poverty, overall. Of children (age 0-17), 75% are in poverty.

This data is likely an underrepresentation for several reasons. First, the Federal Poverty level does not take into account geographic variation in costs of living. For example, the 2020 Federal poverty line for a family of four is \$26,200, but research by the nonprofit Tipping Point Community indicates the poverty threshold for the Bay Area should be set higher—closer to \$40,000—and that the bar for true economic security is much higher—closer to \$75,000.¹⁶⁵

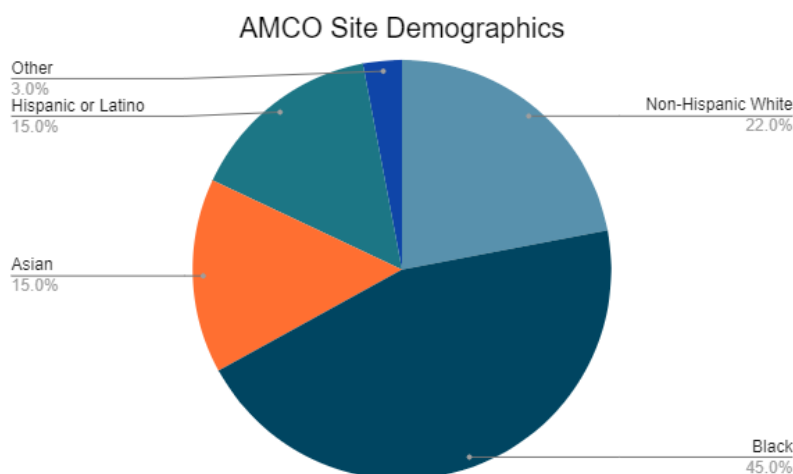


Figure 11: Population by Race/Ethnicity

Second, the census data does not appear to include the homeless community living under the overpass south of the AMCO site. This is particularly important because the contaminated groundwater below the site flows primarily south and southwest, exactly in the direction of the encampment.

¹⁶⁴ U.S. Census Bureau, “American Community Survey: 2015-19.”

¹⁶⁵ Tipping Point Community, UC Berkeley, and Othering & Belonging Institute, “Taking Count: A Study on Poverty in the Bay Area.”

History of Contamination

Contamination by AMCO Chemical Corp.

AMCO Chemical Corp. operated at the site as a chemical distribution facility from the early 1960 to 1989. According to a 2018 EPA report,¹⁶⁶ “The business accepted a wide range of chemicals in bulk and repackaged and distributed chemicals in smaller containers for resale. Bulk chemicals were delivered via a rail spur located at the central area of the Site near the 3rd Street entrance, and the facility stored the chemicals in drums and tanks located throughout the Site.” In 1988, DTSC issued violations related to leaking hazardous waste in deteriorated containers, as well as unlabeled containers. This led to AMCO vacating the site in 1989. Chemicals handled at the site can be broadly categorized as:¹⁶⁷

- Chlorinated organic solvents (TCE, PCE, DCE, VC)
- Non-chlorinated organic solvents
- Petroleum hydrocarbons
- Organochlorine pesticides
- Organophosphate pesticides
- Ethylene glycol
- Other glycol products
- Creosote
- Inorganic salt

Many chemicals within these classes are known as volatile organic compounds (VOCs). These pose a particularly high risk of vapor intrusion (see the prior section of this report describing how sea level rise heightens risks posed by buried contamination). VOCs have been the primary target of the EPAs remediation efforts.

¹⁶⁶ Oneida Total Integrated Enterprises, “AMCO Chemical Superfund Site.”

¹⁶⁷ Department of Toxic Substances Control, “EnviroStor: AMCO Chemical.”

Site Remediation Timeline

Since becoming a Superfund site in 2003, the lead remediation agency for the site has been the EPA, with involvement of DTSC and the RWQCB.

1988

- Aug. 4 DTSC inspected the site
- Aug. 8 DTSC issued “an order to Fence and Post the site”

1989

- Feb. 24 DTSC issued a “Report of Violation... to correct violations related to leakage of hazardous waste from piping and containers; storage of hazardous waste in deteriorated or otherwise corroded conditions; and unlabeled waste containers.”
- Dec. AMCO Chemical Corp. no longer occupied the site.

2004

- July 1 Site was listed on the National Priorities List (most urgent among Superfund)

2016-17

- Feb-Jan First remediation attempt using in-situ thermal heating. The system was not effective and stopped in December 2017 with VOCs, petroleum, LNAPL, VOCs, & SOCs remaining the soil and groundwater beneath the site.

2018

- Sept. Jacobs releases Risk Assessment Technical Approach for use in EPA’s upcoming Non-Time Critical Removal Action report¹⁶⁸
- Oct. EPA releases Non-Time Critical Removal Action report (Oneida)¹⁶⁹

2019

- Feb-Mar Second remediation attempt using in-situ chemical oxidation (injection of sodium permanganate into depths 8 to 15 ft.) This process was ended quickly upon finding that concentrations of some contaminants were increasing: cis-1,2 DCE (by 160% to 3,846%), total petroleum hydrocarbons (both diesel and gasoline).

Updated Aug. 13, 2020 by Project Manager, Lynn Nakashima, DTSC
Source: DTSC EnviroStor database, AMCO site webpage¹⁷⁰

¹⁶⁸ Paik and Shelton, “Risk Assessment Technical Approach: AMCO Chemical Superfund Site, Oakland, California.”

¹⁶⁹ Oneida Total Integrated Enterprises, “AMCO Chemical Superfund Site.”

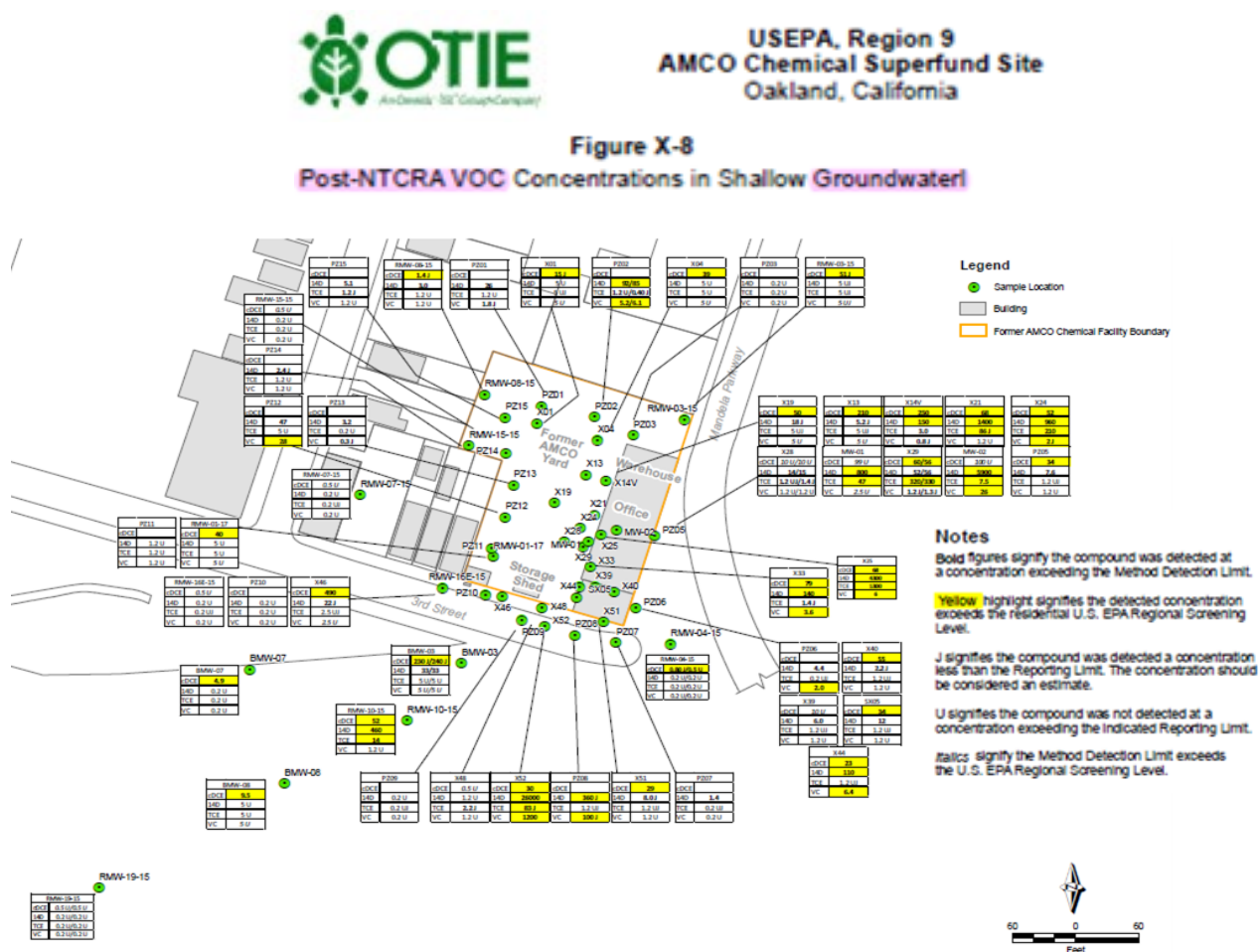
¹⁷⁰ Department of Toxic Substances Control, “EnviroStor: AMCO Chemical.”

Failures of EPA and DTSC Remediation Efforts

The following sections describe failures specific to the AMCO site. Other issues (shared by the other sites profiled in this report as well as other environmental injustices associated with hazardous waste sites in general) are discussed in the following section, “Commonalities.”

Failure to achieve safe chemical concentrations

The remediation agencies’ most glaring failure is the inability to remove the hazardous chemicals at the AMCO site. Throughout the second remediation attempt, the EPA measured chemical concentrations in groundwater, soil, and four nearby residential homes (crawlspace air). Data presented in the EPA’s project report¹⁷¹ show that concentrations (of dozens of chemicals, including the VOCS TCE, PCE, DCE, and VC) remained above safe levels through the end of the remediation work (see Figure 12).¹⁷²



¹⁷¹ Oneida Total Integrated Enterprises, “AMCO Chemical Superfund Site.”

¹⁷² Oneida Total Integrated Enterprises.

Perhaps the most concerning failure is the EPA’s apparent intention to exclude residential homes from contamination monitoring and remediation work. To inform the subsequent remediation work conducted in February-March 2019 (in-situ chemical oxidation as described in the October 2018 report prepared by Oneida¹⁷³), the EPA commissioned environmental consultant firm, Jacobs, to prepare the “conceptual site model” (CSM). The purpose of a CSM is to determine the sources of potentially dangerous chemicals, their environmental fate and transport (i.e., the extent to which they partition to air, soil, and water), human exposure pathways (e.g., VOCs in groundwater that seep into homes as vapors), and resulting human health risk. As noted in a comment letter from DTSC scientist Lynn Nakashima to EPA site manager Kelly Manheimer, the CSM constructed by Jacobs¹⁷⁴ failed to “acknowledge off-site receptors and address their potential exposure to site-sourced contaminants,”¹⁷⁵ with “off-site receptors” being the people living nearby.

Further evidence of the EPA’s intention to exclude residents from its risk assessments and remediation responsibility is its conclusion that the site has achieved “Human Exposure Under Control.”¹⁷⁶ The EPA’s decision on this performance metric is listed on the AMCO site’s Superfund profile webpage,¹⁷⁷ and indicates that “there are no unacceptable human exposure pathways.” Given that, according to last measurements reported in EPA documentation,¹⁷⁸ the VOC concentrations in both *nearby homes* and the *groundwater* beneath the site (both measured in August 2018) were above safe levels, the EPA could only come to this conclusion by excluding nearby homes from its risk assessment.

Failure to warn residents of risk

The failure to remediate is exacerbated by the EPA’s misleading and incomplete communication with the community regarding the severity of the health risks. The EPA sent emails to community members to announce remediation work planned and in progress (see Appendix for email), but they neglect to alert residents that concentrations of VOCs and petroleum compounds in groundwater are still unsafe (Figure 8).

The email mentions planned soil and groundwater testing in September of 2019. To the authors’ knowledge at the time of writing this report, there are no documented results from that testing or indications that it occurred at all. Similarly, there are no documented actions taken to protect nearby homes against vapor intrusion, so there is no reason to believe the unsafe concentrations measured in March have declined.¹⁷⁹

¹⁷³ Oneida Total Integrated Enterprises.

¹⁷⁴ Paik and Shelton, “Risk Assessment Technical Approach: AMCO Chemical Superfund Site, Oakland, California.”

¹⁷⁵ Nakashima, “Comments on Risk Assessment Technical Approach,” November 1, 2018.

¹⁷⁶ “AMCO Chemical Superfund Site Profile.”

¹⁷⁷ “AMCO Chemical Superfund Site Profile.”

¹⁷⁸ Oneida Total Integrated Enterprises, “AMCO Chemical Superfund Site.”

¹⁷⁹ Manheimer, “FW: AMCO Update,” July 2, 2019.

Furthermore, some residents near the AMCO site rent. While visiting the AMCO site and surrounding area, one of the authors of this report (Emily Kraemer) spoke with the residents living in the closest home to the site (the home furthest southwest in Figure 8), a woman and her husband. When asked about the AMCO site, they said they had not heard anything about either contamination or remediation work, explaining that they rent and their landlord had not mentioned anything.

Health Impacts

When inhaled over many years, VOCs tend to cause respiratory issues, including asthma and bronchitis as well as other issues, including cancer. Transportation-related air pollution is particularly acute in the area surrounding the AMCO site, given this western-most part of West Oakland is the most proximate to railways and shipping yards, in addition to Interstate 880. Respiratory irritants have compounding effects, thus creating a situation where residents are extraordinarily vulnerable to additional air pollutants and even “safe” levels of VOCs in people homes can yield significant harm, contributing to an already high asthma rate (Figure 13) and lower life expectancies (Figure 14).¹⁸⁰

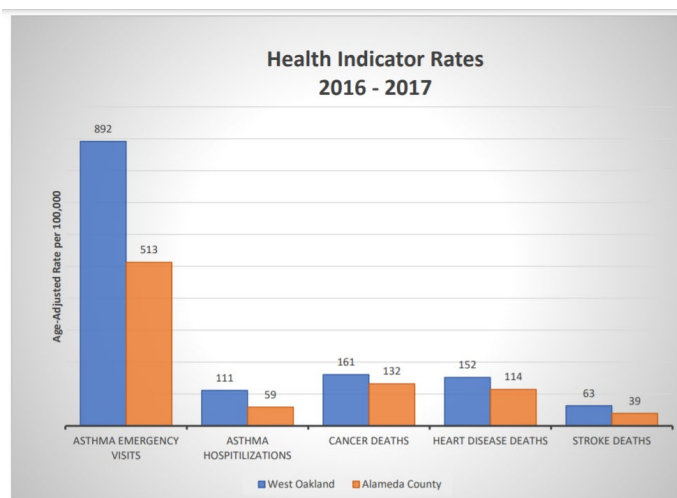
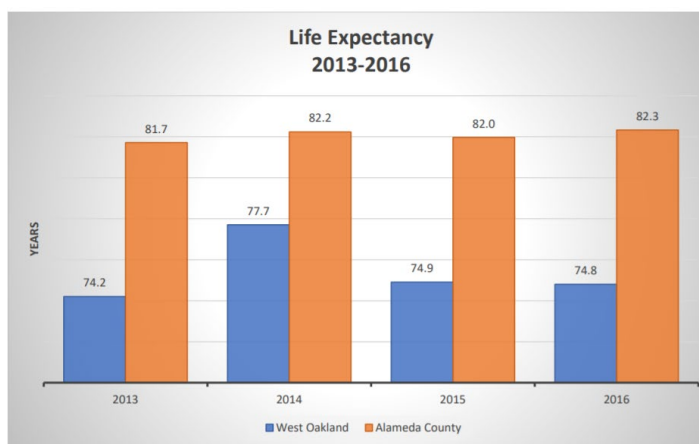


Figure 13: People living in West Oakland experience higher rates of health issues, including asthma



Source: ACPHD CAPE, with data from Alameda County vital statistics files, 2000-2017

Figure 14: People living in West Oakland tend to have shorter life expectancies, in part due to high levels of pollution in their neighborhoods

¹⁸⁰ West Oakland Environmental Indicators Project (WOEIP) and Bay Area Air Quality Management District, “Owning Our Air - Volume 2.”

Commonalities Shared Across Cases of Environmental Injustices Associated with Hazardous Waste Sites

How Contamination Lands in Communities: Zoning and Race

For decades now, academics, journalists, and activists working in the field of Environmental Justice have shown how economic zoning leads to environmental racism.¹⁸¹ The case studies presented here show how zoning laws that deem non-White neighborhoods as industrial zones, cities concentrate hazardous waste, adverse health impacts (e.g., asthma), and poverty tend to concentrate in communities of color (Figures 15 and 16).

In many ways, these case studies exemplify the idea of “codified neglect.” Liza Gross describes how Tulare County, in its water systems development plan, identified “non-viable communities” to describe communities that would not be connected to treated drinking water distribution.¹⁸² Similarly, neighborhoods first labeled “blighted” or “slum” were later zoned by city planners as “industrial” or “mixed-use industrial/residential.”¹⁸³ These labels describe the neighborhoods in which companies and government have been allowed to locate industrial waste sources near where people live, work, and play.

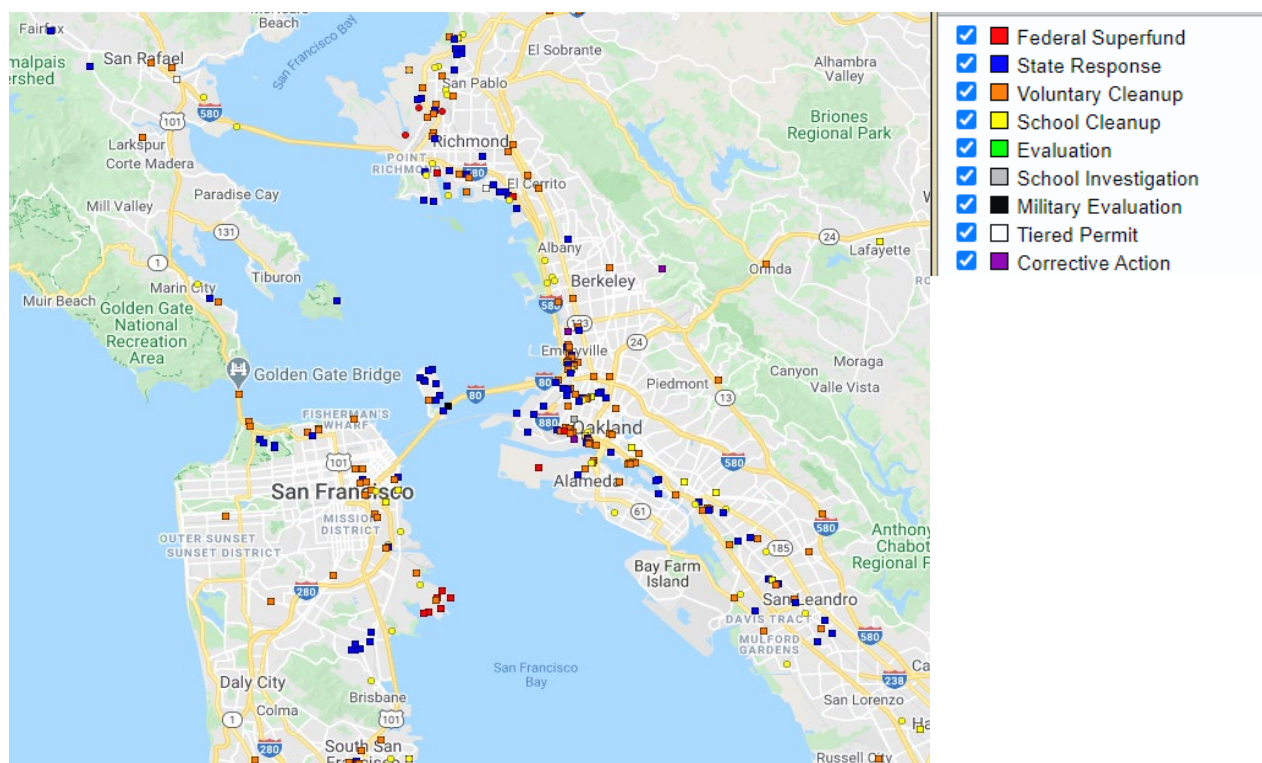


Figure 15: Hazardous waste sites in the Bay Area

¹⁸¹ Holtzapple, “How Economic Zoning Promotes Environmental Racism.”

¹⁸² Gross, “No Beba El Agua. Don’t Drink the Water.”

¹⁸³ Madrigal, “Oakland - How the Bay Was Built.”

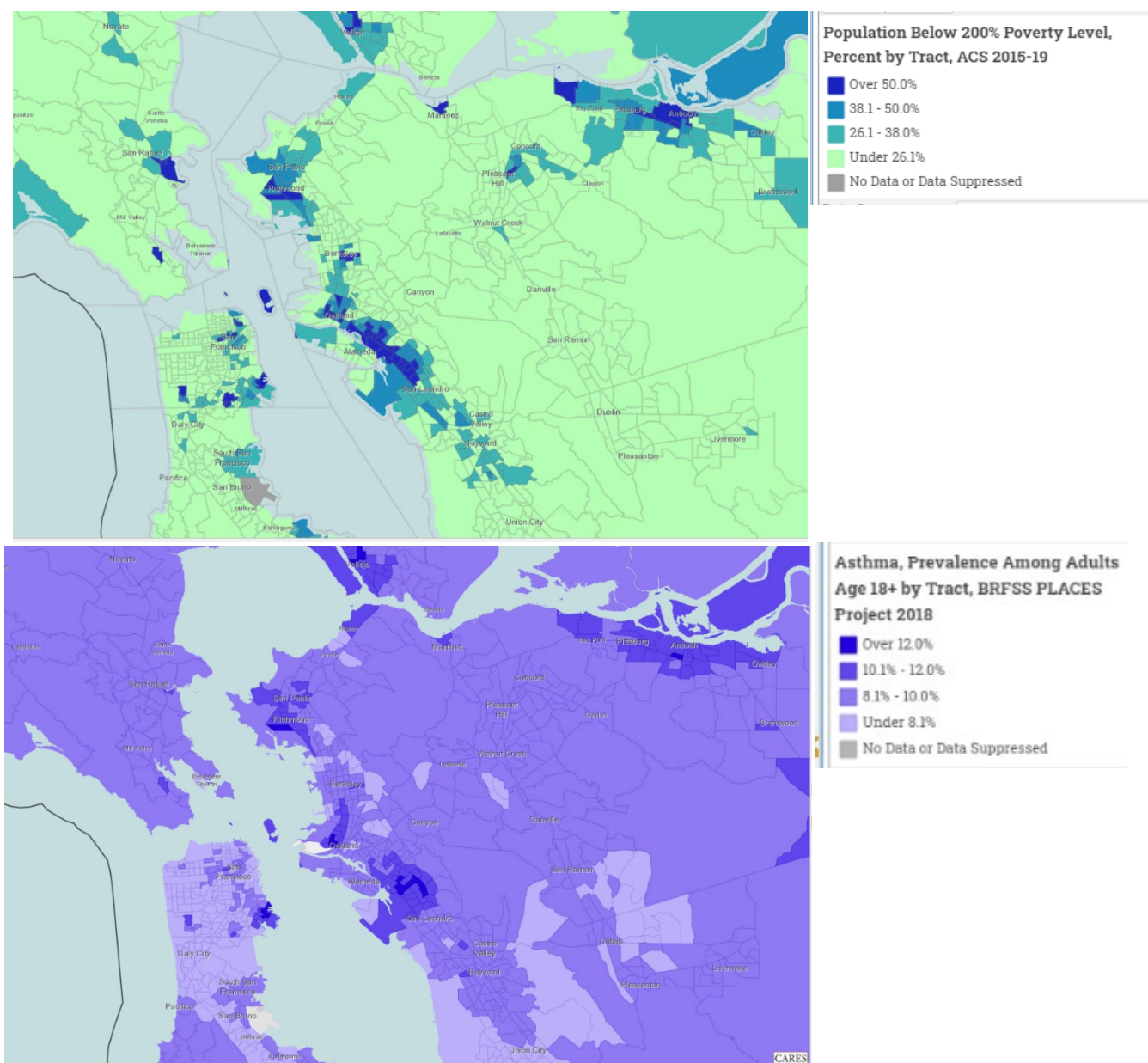


Figure 16: Poverty (top) and asthma rates in the Bay Area

How Hazardous Waste Remains in Communities of Color: Technocratic Approaches to Environmental Justice

In her assessment of how to create environmentally just technologies, Gwen Ottinger articulates a key challenge shared by these case studies. “Technocratic approaches to environmental policy, including quantitative risk assessment, have been shown to exclude affected communities from decision-making...by structuring policy discourses so as to devalue their contributions.”¹⁸⁴

The EPA’s and DTSC’s approaches to hazard risk assessment and remediation reflect exactly this mindset (see Figure 17),¹⁸⁵ which Donna Riley also discusses with respect to how it stands in the way of environmental justice.¹⁸⁶ Riley argues that engineers have a “tendency toward ‘brute force methods of problem solving and an exclusive focus on the calculated and measured solution,” while discounting the credibility of other forms of knowledge, derived through human interactions. Riley shares a joke (“The Church Steeple”) commonly told among engineers to illustrate this:

“An engineer and a sociologist were tasked with finding the height of a church steeple. The engineer measured the angle to the top of the steeple and calculated the height using trigonometry. Then, to check the estimate, the engineer climbed to the top of the steeple, lowered a string until it touched the ground, climbed back down and measured the length of the string. The engineer compared the measurement to the estimate, calculated the standard error, and drafted a report documenting the methods and results. The sociologist bought the sexton a beer in the local pub and he told her how high the church steeple was.”¹⁸⁷

¹⁸⁴ Ottinger, “Environmentally Just Technology.”

¹⁸⁵ “Superfund Remedy and Non-Time Critical Removal Action (NTCRA) Decision Documentation,” United States Environmental Protection Agency, accessed May 14, 2021, <https://www.epa.gov/superfund/superfund-remedy-and-non-time-critical-removal-action-ntcra-decision-documentation#remedial>.

¹⁸⁶ Riley, “Engineering and Social Justice.”

¹⁸⁷ Riley, 36.

In the same way, the highly technical and quantitative ways in which regulators seek to prove human health risk through complicated conceptual site models and expensive field sampling campaigns seems much more difficult than simply listening to residents and community activists, such as Greenaction and WOEIP.

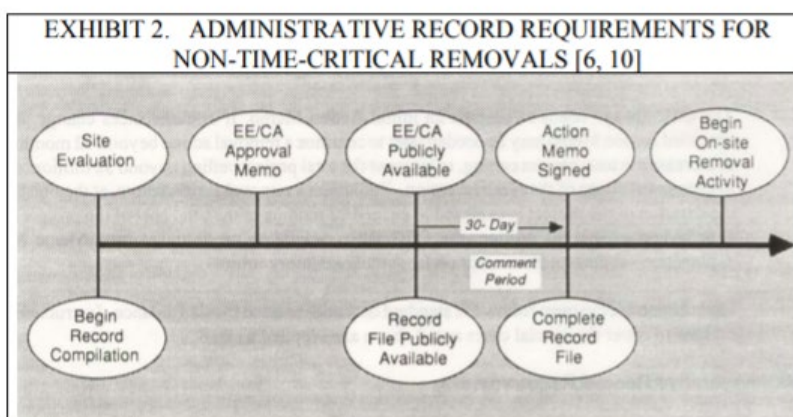


Figure 17: The EPA’s process for site remediation, including pre and post evaluation of risk

The EPA’s Hazard Ranking System and Superfund-Site Designation

The speed and extent to which sites are remediated is greatly impacted by whether the EPA designates them as Superfund Sites.

The EPA has a convoluted, multi-phase administrative process for site remediation. Each step and nested sub-steps are laden with quantitative modeling and unnecessarily lengthy terminology and acronyms. The EPA currently classifies BVHP, the AMCO site, and a site within the Zeneca area as Superfund Sites (see Table 1). TI seems to have qualified for Superfund designation, but it is not clear why it is no longer listed as such, as covered by the SF Chronicle:

“The process of listing a Superfund site begins with the EPA’s Hazard Ranking System, which measures the threat to human health and the environment on a 100-point scale. A score above 28.5 qualifies that place for a Superfund designation, which would make cleanup a federal priority. In 1991, the EPA calculated a hazard score for Naval Station Treasure Island, the base that included all of Treasure Island... The base’s score was 51.78, the new documents show, almost double the threshold for Superfund consideration and slightly higher than the score for the Hunters Point Naval Shipyard in the southeast corner of San Francisco, which was named a Superfund site in 1989... Environmental advocacy groups said the decision led to a dysfunctional and delayed cleanup, making the process less transparent and leaving thousands of Treasure Island residents in the dark for years about contamination near their homes. In 2007, when Navy contractors started to discover radioactive objects across the island that weren’t supposed to be there, the EPA officially remained on the sidelines without ever fully explaining why.”¹⁸⁸

¹⁸⁸ Fagone and Dizikes, “SF’s Treasure Island, Poised for Building Boom, Escaped Listing as Superfund Site.”

Table 1. The EPA assesses potential Superfund sites using its Hazard Ranking System, which calculates the treat to human and environmental health. Out of a maximum of 100, a score above 28.5 qualifies a site for Superfund designation.

Site Name	City	Site EPA ID	Year of Superfund Status Decision	EPA Hazard Score*	National Priorities List
EPA Publicly Acknowledges Superfund Status:					
Hunters Point Naval Shipyard	San Francisco	CA1170090087	1989	48.77	Delisted
Liquid Gold Oil Corp (adjacent to Zeneca)	Richmond	CAT000646208	1983	43.32	Delisted
AMCO	Oakland	CA0001576081	2003	50.00	Yes
Superfund Status is Ambiguous:					
Treasure Island	San Francisco	CA0902776	1991	51.78	n/a

Thus, the process by which sites and the affected communities are prioritized is inconsistent and likely swayed by outside forces, including political and economic. Bullard and Wright observed similar instances of regulators failing to protect communities from hazardous contamination. In their analysis of Black communities in the south faced with industrial pollution, they argue that because the EPA’s ten regions “operate as nearly autonomous subagencies,” they can easily make deals and compromises with state and local governments, which has resulted in “many... bad facility siting and permitting decisions” disproportionately expose people of color to hazardous contamination and pollution.¹⁸⁹

Reliance on Too-High Chemical Safety Limits

As the lead oversight agency on Superfund Sites, the EPA determines what qualifies as unsafe levels on a chemical-by-chemical basis via conceptual site models (as described in the AMCO site case study). These “site specific screening levels” claim to be based on quantitatively calculation of the probability that a person will be exposed to a contaminant and that it will cause adverse health impacts. Remedial action is only deemed necessary if chemical concentrations are measured and found to exceed the decided limit.

However, these modelling procedures do not (and cannot) account for exposure to other contaminants (e.g., air pollution from highways, shipping ports, and railways; other hazardous waste sites). Despite consensus on (and admission to) the compounding impact on a person’s health, they calculate health risk as the sum of its parts.

¹⁸⁹ Bullard and Wright, *The Wrong Complexion for Protection*, 125.

Furthermore, even the most robust monitoring campaign cannot measure all the known and possible contaminants. Even if it could, we have incomplete or nonexistent information about their various health impacts (especially long-term). This is because the dozens of chemicals known to have been released at hazardous waste sites also have innumerable transformation products.

For these reasons, even “safe” levels of contaminants cause health issues, especially for those who are exposed for many years. This is no surprise, given the fact that these levels change often, reflecting realizations that a chemical is more harmful than previously understood. Remedial action that seeks only to reach the currently decided safety limit is insufficient.

Conclusion

Similarities Across the Four Case Studies

BVHP and TI share a history of contamination defined by the involvement of the U.S. Navy. Although the Navy has instituted cleanup efforts, these efforts fall far short of the line necessary to redevelop the sites. The cleanup efforts have their unique issues, but leave the community living in the areas in danger from contamination. Both sites have received significant public and media interest, and community groups are working to ensure the sites receive proper cleanup efforts in order to protect their residents.

The Zeneca site in Richmond and the AMCO site in West Oakland were both contaminated by private companies. They have undergone cleanup efforts led by government agencies, but both cleanup efforts do not adequately protect nearby residents from the contaminants of the site. These sites have not received as much public and media interest, leaving the residents at the mercy of the government's inadequate response.

All four of the sites in this report are quite vulnerable to the effects of climate change. Additionally, climate change will exacerbate the contamination of the sites, rendering much of the historical cleanup efforts useless due to rising groundwater and other effects.

Synopsis and Next Steps

The Bay Area has many contaminated sites, including the sites mentioned in this report. However, most of them have been ignored and swept under the rug, leaving vulnerable communities at risk for health effects caused by the toxic environment.

Climate change will exacerbate environmental injustices at the sites studied in this report. Sea level rise will increase flood risks and cause groundwater rise. Groundwater rise will bring its own host of issues, including destabilizing structures, putting them at risk of collapse during an earthquake, and activating contaminants buried beneath the surface, unearthing threats buried in the past.

Lastly, these case studies bring up many questions about contaminated sites and environmental justice issues in general—who is responsible in remediating the site? What agency has the authority to approve? How does this issue tie with other issues within the Bay Area, such as the housing crisis? Further research and engagement must be done to learn more about these issues and to develop effective solutions preventing similar problems in the future.

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Appendix

Email from EPA site manager to residents living near the AMCO Superfund Site

From: [Manheimer, Kelly](#)
To: [Manheimer, Kelly](#)
Subject: FW: AMCO Update
Date: Tuesday, July 2, 2019 5:26:04 PM

Dear neighbors and interested parties,

EPA is sending this e-mail to the AMCO community e-mail list to provide an update on upcoming environmental cleanup activities.

Background

In March, EPA completed an initial phase of testing a cleanup technology called “in-situ chemical oxidation” or “ISCO”. The purpose of the testing (also referred to as a “treatability study,”) is to evaluate if ISCO can be successfully used to address a localized area of groundwater contamination underneath the warehouse. The contaminants in the groundwater include the solvents “tetrachloroethene” or “PCE” and “trichloroethene” or “TCE” from historic activities at the site. Groundwater samples collected in May 2019, indicated that the contaminant concentrations were lower in some of the wells underneath the warehouse; however, concentrations of petroleum products (gasoline and diesel) in the groundwater were higher than expected.

Next Steps

Because of the higher petroleum constituents, we have decided to hold off on additional ISCO injections for the time being, and will take another round of sampling towards the end of July/early August 2019 to further characterize the impacts of the first ISCO injection. As previously, we will be able to work around ongoing operations on the site and no interruption of work should occur.

We will send another update in August or September after we receive the results of the latest round of sampling and we have determined a path forward. This update will include a schedule for additional sampling of the groundwater and our plans for completing the treatability testing. If you have any questions, please contact me at 415-972-3290 or by e-mail to manheimer.kelly@epa.gov. You may also contact EPA’s Community Involvement Coordinator, Jackie Lane, at (415) 972-3236 or by e-mail to lane.jackie@epa.gov.

We greatly appreciate your cooperation. Have a wonderful day!

Regards,

Kelly Manheimer

Section Manager, CA Sites II
EPA Region 9 Superfund
415-972-3290